Supplementary material
Unit-Grouped Evaluation of Power-On Built-In Test Classifiers for Airborne Actuators: Critical Misses, False Alarms and Independent-Unit Evidence

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S1. Supplementary analyses

This supplementary file reports analyses that support, but do not change, the conclusions of the main text: exploratory nested tuning and its selection run, seed stability, per-class recall in a single seed, and sensitivity to equivalence margins and fixed hyperparameters. It also documents the feature inventory, auxiliary F06 checks and excluded earlier analyses. It also gives, verbatim, the data-quality audit, methodological details and results that the main text summarises (Sections S1.23, S1.24 and S1.25).

Abbreviations. ARP: Aerospace Recommended Practice; BCa: bias-corrected and accelerated; CMR: critical miss rate; CRC: conformal risk control; CSV: comma-separated values; CV: coefficient of variation; DR: downward risk; FA0: false-alarm rate on normal cycles; FWER: family-wise error rate; ID: identifier; IQR: interquartile range; MAD: median absolute deviation; PBIT: power-on built-in test; RF: Random Forest; RMS: root mean square; SE: severity error; SMOTE: Synthetic Minority Over-sampling Technique; SMOTEWB: SMOTE with boosting; SN: serial number; TPE: tree-structured Parzen estimator; XLSX: Microsoft Excel workbook format. Labels E1 to E16 identify the archived analysis protocols referred to in the main text and in this file.

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S1.1. Exploratory CMR-Targeted Hyperparameter Optimisation

This auxiliary analysis examines the historical CMR-targeted search under nested evaluation (Section 4.6). It supplies model configurations for the descriptive operating-point comparisons, but is not an independent contribution or a controlled test of the search objective. Table S1 reports the nested results; the values of the selection run (S1.2) are optimistic, because the same folds select and evaluate the configuration.

Table S1: Nested validation of LightGBM tuning

LightGBM configuration	**CMR [95% CI]**	**FA0**	**Downward risk**	R_3	**Macro-F1**
Untuned	0.070 [0.018; 0.147]	0.371	0.165	0.825	0.390
F1-tuned (nested)	0.065 [0.025; 0.112]	0.410	0.189	0.795	0.389
CMR-tuned (nested)	0.030 [0.009; 0.057]	0.790	0.094	0.885	0.326
CMR-tuned, selection-run parameters	0.030 [0.005; 0.066]	0.711	0.111	0.875	0.354

Note: Five outer and five inner grouped folds, 54 units. CMR has 95% intervals from resampling serial numbers. The last row applies the selection-run parameters, which saw all folds, to the outer folds.

Under the historical search procedure, the nested CMR of the CMR-tuned configuration is 0.030 (descriptive 95% interval 0.009 to 0.057), against 0.020 on the selection folds and 0.009 averaged over the best inner values. Relative to the untuned model (0.070), the reduction is 0.040 (descriptive 95% interval 0.000 to 0.095). These percentiles describe sensitivity to the sampled units and do not test a reduction. Relative to the F1-tuned configuration (0.065) the reduction is 0.035 (descriptive 95% interval 0.009 to 0.064), whereas the F1-tuned configuration does not reduce CMR relative to the untuned model (0.005; descriptive 95% interval −0.024 to 0.043). The observed relative reduction between search procedures is 54% (0.065 to 0.030), against 72% (0.072 to 0.020) on the selection folds. It cannot be attributed to the objective alone because budgets, boost spaces and feasibility rules differ. The reduction costs 0.064 in macro-F1 (descriptive 95% interval 0.016 to 0.110) and raises FA0 from 0.371 to 0.790. The parameters of the selection run, applied without nesting, give the same pooled CMR (0.030; difference 0.000, descriptive 95% interval −0.012 to +0.018). This reused-parameter comparison does not establish absence of selection optimism or equivalence. The searches selected λ_{sev3} between 10.8 and 17.7 (bound 20) and reached the lower bound of the learning rate in some folds, so the search space was binding. Only LightGBM was assessed with nested selection.

S1.2. Selection Run of the CMR-Targeted Search

The selection run is reported because the nested validation (Section S1.1) uses its parameters as a comparator and because its values show how optimistic non-nested selection is. It used the search of Section 4.6 on the full training partition, with the severity-boost weights of Eq. (S2). On the full training partition, 67 of the 100 trials satisfied the constraint. The best trial reached a selection-fold CMR of 0.020 with $\lambda_{\text{sev3}} = 19.5$ (near the upper bound of the search range) and $\lambda_{\text{f05}} = 2.9$. An F1-tuned configuration (50 trials maximising the mean macro-F1) is the comparator. The values observed on the selection folds are optimistic because the same folds select and evaluate the configuration.

On these folds, the CMR-tuned configuration reduces the CMR from 0.072 to 0.020 and the downward risk from 0.164 to 0.107, at a cost of 7.4 percentage points in macro-F1 (0.352 against 0.427); F05 and F06 recall are unchanged (both 0.667).

S1.3. Seed Stability

The comparisons of the main text pair ten seeds, so the variability of the metrics across seeds must be known to read them and to set the equivalence margins (Section 4.2). Table S2 shows that the F06 recall (coefficient of variation 40.5%) and the CMR (18.1%) vary more across seeds than the downward risk (4.3%) and the balanced accuracy (6.6%). The standard deviations of the CMR (0.014) and of the macro-F1 (0.031) are the reference for the margins of 0.02 and 0.03.

Table S2: Stability across random seeds

Metric	Mean	Std	Min	Max	CV%
CMR	0.078	0.014	0.056	0.102	18.1
Downward Risk	0.176	0.008	0.164	0.189	4.3
Macro F1	0.386	0.031	0.343	0.427	8.0
Balanced Acc	0.504	0.033	0.460	0.552	6.6
F06 Recall	0.483	0.196	0.250	0.667	40.5
Severity-3 recall	0.757	0.082	0.624	0.833	10.9

Note: Ten seeds in five-fold StratifiedGroupKFold validation, LightGBM F1-tuned configuration, selection folds. Std: standard deviation; CV%: coefficient of variation.

S1.4. Per-Class Recall in a Single Seed

The aggregate metrics of Section 5.3 do not show which classes drive the errors, so the per-class recall of a single seed is reported for description only. Figure S1 shows LightGBM (246 descriptors, F04/F05/F06 augmentation to 160 cycles, seed 42): F03 (82%), F01 (78%) and N01 (64%) are recognised most often, F02 (41%, n=259) is the largest contributor to misclassification given its class size, F04 reaches 38%, F05 7% and F06 33% (two of six training rows), and N02 0%.

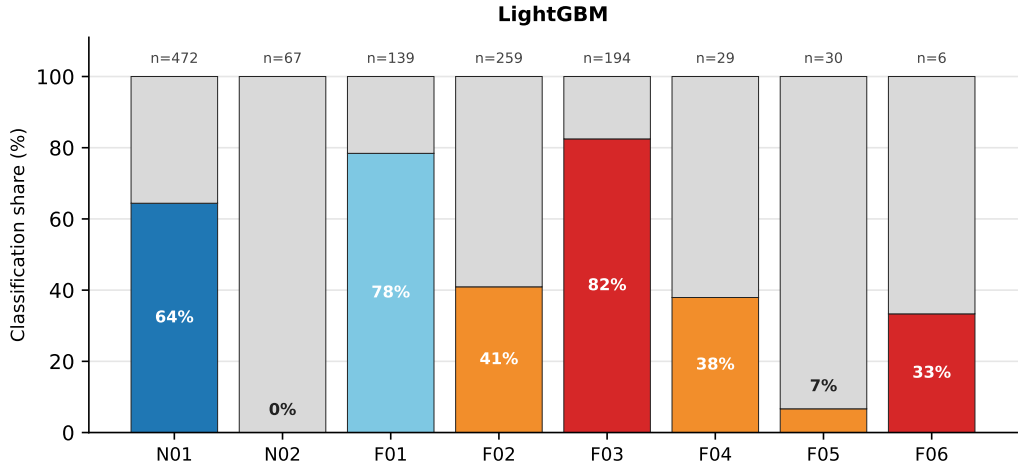


Figure S1: Per-class correct (coloured) and incorrect (grey) classification percentages for the current LightGBM grouped five-fold configuration (246 features, F04/F05/F06 augmentation target = 160, noise = 1%). Bar colours reflect the ARP4761 severity level: blue = Severity 0 (Normal), light blue = Severity 1, orange = Severity 2, red = Severity 3 (Critical).

S1.5. Sensitivity to the Equivalence Margin

This subsection documents the historical equivalence and difference criteria of the retrospective correction. The main text no longer uses them, because no procedure was calibrated for paired CMR differences (Section S1.8).

Because the margins are choices of the authors, the historical conclusions of equivalence are checked against other margins. The margins of Section 4.2 are tolerances of the order of the seed-to-seed standard deviation, not operational limits: a margin of 0.02 in CMR corresponds to four of the 200 severity-3 cycles of the training partition, which come from 14 units. Table S4 gives historical unadjusted 90% intervals for the paired differences and the smallest symmetric

margin $\delta_{\min}$ containing each; the unadjusted interval criterion holds at any larger margin. These historical 90% intervals are unadjusted exploratory summaries and do not support family-level equivalence. At half the historical margins, only the Gaussian-versus-none macro-F1 contrast among the formerly designated equivalences passes the unadjusted criterion. The comparisons between classifier families ($\delta_{\min}$ 0.060 and 0.080) and between F246 and F18 in CMR (0.092 for LightGBM, 0.117 for Random Forest) failed the historical containment criterion at every margin of the grid. The downward-risk interval for LightGBM with F246 against F18 was contained only from a margin of 0.023, above the historical 0.02. Neither outcome establishes population equivalence or non-equivalence.

Table S3 reports the retrospective correction using 10,000 paired SN resamples. The feature-reduction family uses 98.75% intervals for eight hypotheses; its CMR contrasts need margins greater than 0.0265 (LightGBM) and 0.0246 (Random Forest), so neither met the historical criterion at 0.02. Four of the eight endpoints met the historical containment criterion, but neither model met it jointly for CMR and FA0. The three LightGBM imbalance contrasts used 96.67% intervals and met that historical criterion on their specified endpoints; none passes at half the adopted margin. These family definitions were made after examining the historical results and are not prospective confirmation.

S1.6. Sensitivity to the Fixed Hyperparameters

The hyperparameters of Section 4.5 were fixed without a prior sensitivity study. To bound the generality of Sections 5.3 to 5.3, one setting at a time was varied around them under the protocol of Section 5.1 (13 LightGBM and 6 Random Forest variations; Table S5). For LightGBM, eight of the 13 historical CMR intervals at 99.23% lay within ±0.02. This describes the former interval criterion, not established equivalence. The historical Holm-adjusted tests are superseded by the negative coverage assessment (Section S1.8); no current inference is drawn from their values. Fewer trees (100) raises FA0 by 0.027 (descriptive 95% interval 0.019 to 0.036) and lowers the downward risk by 0.007 (0.001 to 0.013). The row-bagging contrast is descriptive; it does not establish absence of a CMR effect. The base setting of this sensitivity analysis does not use row bagging. For Random Forest, in contrast, the minimum leaf size alone moves the CMR from 0.112 (size 1) to 0.071 (base, size 2) and 0.041 (size 5), with FA0 moving the other way (0.407, 0.503, 0.685), consistent with a change of operating point; operational superiority is not established.

Table S3: Historical multiplicity-adjusted bootstrap criteria

Comparison	Metric	Interval level	Corrected interval	$\delta_{\min}$	Equivalent
RF − LightGBM	CMR	97.50%	[-0.108; +0.070]	0.108	no
RF − LightGBM	FA0	97.50%	[-0.017; +0.110]	0.110	no
RF − XGBoost	CMR	97.50%	[-0.081; +0.073]	0.081	no
RF − XGBoost	FA0	97.50%	[-0.069; +0.030]	0.069	no
LightGBM: F246 − F115	CMR	98.75%	[-0.027; +0.013]	0.027	no
LightGBM: F246 − F115	FA0	98.75%	[-0.036; +0.015]	0.036	no
LightGBM: F246 − F115	DR	98.75%	[-0.013; +0.016]	0.016	yes
LightGBM: F246 − F115	Macro-F1	98.75%	[-0.006; +0.039]	0.039	no
RF: F246 − F115	CMR	98.75%	[-0.025; +0.017]	0.025	no
RF: F246 − F115	FA0	98.75%	[-0.017; +0.020]	0.020	yes
RF: F246 − F115	DR	98.75%	[-0.005; +0.013]	0.013	yes
RF: F246 − F115	Macro-F1	98.75%	[-0.012; +0.003]	0.012	yes
P1: weights	CMR	96.67%	[-0.016; +0.011]	0.016	yes
P2: Gaussian − none	Macro-F1	96.67%	[-0.009; +0.017]	0.017	yes
P3: Gaussian − SMOTE	Macro-F1	96.67%	[-0.024; +0.015]	0.024	yes

Note: 10,000 paired cluster resamples, 54 SN and ten fixed seeds. Interval levels are $1 - 2(0.05)/m$: 97.50% for four family contrasts, 98.75% for eight feature endpoints and 96.67% for three imbalance contrasts. Analytical margins: 0.02 for CMR/DR, 0.03 for FA0/macro-F1. P1 compares product with frequency weights in CMR; P2 and P3 compare Gaussian augmentation with none and SMOTE in macro-F1. No paired resample lacked an endpoint denominator. "Equivalent" denotes the historical interval criterion only: the subsequent coverage assessment did not validate a CMR/FA0 procedure across all scenarios, and did not assess DR/macro-F1.

Table S4: Historical interval-containment margins

Comparison	Metric	Δ	90% interval	$\delta_{\min}$	Adopted margin	Equivalent at 0.5 / 1.5 / 2.5×
Families						
RF − LightGBM	CMR	−0.010	[−0.080; +0.049]	0.080	0.02	no / no / no
RF − XGBoost	CMR	−0.005	[−0.060; +0.050]	0.060	0.02	no / no / no
Features, LightGBM						
F246 − F115	CMR	−0.002	[−0.015; +0.008]	0.015	0.02	no / yes / yes
F246 − F115	DR	+0.002	[−0.008; +0.011]	0.011	0.02	no / yes / yes
F246 − F115	F1	+0.013	[−0.001; +0.029]	0.029	0.03	no / yes / yes
F246 − F18	CMR	−0.049	[−0.092; −0.003]	0.092	0.02	no / no / no
F246 − F18	DR	−0.003	[−0.023; +0.019]	0.023	0.02	no / yes / yes
F246 − F18	F1	+0.008	[−0.007; +0.026]	0.026	0.03	no / yes / yes
Features, RF						
F246 − F115	CMR	−0.002	[−0.015; +0.010]	0.015	0.02	no / yes / yes
F246 − F18	CMR	−0.080	[−0.117; −0.041]	0.117	0.02	no / no / no
Imbalance, LightGBM						
P1	CMR	−0.002	[−0.013; +0.008]	0.013	0.02	no / yes / yes
P2	F1	+0.005	[−0.006; +0.014]	0.014	0.03	yes / yes / yes
P3	F1	−0.002	[−0.019; +0.012]	0.019	0.03	no / yes / yes

Note: Ten paired seeds and 54 units. RF: Random Forest; DR: downward risk; F1: macro-F1; Δ: paired difference. $\delta_{\min}$ is the smallest symmetric margin containing the historical 90% interval; interval containment does not establish equivalence because coverage was not validated. Adopted margins are 0.02 for CMR/DR and 0.03 for F1. The last column reports equivalence at 0.5, 1.5 and 2.5 times the adopted margin. P1: severity–frequency versus frequency weights, with Gaussian augmentation; P2: Gaussian versus no augmentation, with product weights; P3: Gaussian versus SMOTE, with product weights.

The comparison between families in Section 5.3 therefore holds for the fixed settings, and a different Random Forest setting would change the CMR of that family by more than the differences between families (at most 0.010).

Table S5: Sensitivity to fixed hyperparameters

Model	Setting	CMR	FA0	Macro-F1	CMR minus base [95% CI]
LightGBM	base setting	0.081	0.461	0.378	—
	100 trees	0.074	0.488	0.373	−0.007 [−0.017; +0.001]
	200 trees	0.084	0.464	0.378	+0.004 [+0.000; +0.008]
	800 trees	0.082	0.461	0.378	+0.002 [−0.001; +0.005]
	learning rate 0.02	0.081	0.464	0.378	+0.000 [−0.004; +0.003]
	learning rate 0.10	0.080	0.462	0.378	−0.001 [−0.005; +0.003]
	15 leaves	0.071	0.456	0.385	−0.009 [−0.038; +0.008]
	31 leaves	0.076	0.455	0.386	−0.004 [−0.014; +0.002]
	127 leaves	0.083	0.463	0.378	+0.002 [−0.003; +0.007]
	minimum child samples 2	0.093	0.466	0.371	+0.012 [+0.002; +0.027]
	minimum child samples 20	0.074	0.457	0.382	−0.007 [−0.030; +0.010]
	column subsampling 0.5	0.079	0.461	0.386	−0.002 [−0.009; +0.004]
	column subsampling 1.0	0.085	0.461	0.378	+0.004 [−0.002; +0.011]
	row bagging enabled (0.8)	0.088	0.454	0.384	+0.008 [−0.012; +0.029]
Random Forest	base setting	0.071	0.503	0.359	—
	100 trees	0.073	0.507	0.359	+0.002 [−0.003; +0.008]
	1000 trees	0.070	0.504	0.371	−0.001 [−0.005; +0.003]
	minimum leaf size 1	0.112	0.407	0.371	+0.041 [+0.026; +0.059]
	minimum leaf size 5	0.041	0.685	0.344	−0.030 [−0.078; −0.003]
	features per split $\log_2 p$	0.095	0.478	0.365	+0.024 [+0.003; +0.050]
	features per split $0.3p$	0.065	0.436	0.372	−0.006 [−0.041; +0.022]

Note: 246 descriptors, ten paired seeds and 54 units. One setting is changed at a time. The last column gives the paired CMR difference from the base setting with a 95% interval from resampling serial numbers.

S1.7. False Alarms by Unit

We decomposed the saved historical predictions without refitting: ten seeds for grouped cross-validation (CV) and one for the held-out test. The CV population contains 539 normal cycles from 28 serial numbers, including experiment 1107; the test contains 114 N01 cycles from seven units. N02 occurs in SN 030 and SN 045, with seven and 60 cycles, respectively. Unit summaries combine N01 and N02 within each serial number.

For LightGBM, the three units with the most false alarms contribute 42.0% of alarms in 21.7% of normal-cycle exposure. Removing SN 045 reduces FA0 from 0.461 to 0.410; its contribution does not explain the full CV–test difference. In the F1-tuned test, SN 031 contributes three alarms in four

Table S6: Cycle-weighted and unit-weighted false-alarm rates

Model	Units	Cycle FA0	Unit mean	Unit median	Without SN 045
LightGBM (CV)	28	0.461	0.428	0.407	0.410
Random Forest (CV)	28	0.503	0.466	0.432	0.464
XGBoost (CV)	28	0.524	0.498	0.482	0.477
Gradient Boosting (CV)	28	0.573	0.540	0.542	0.536
F1-tuned (test)	7	0.061	0.264	0.048	–
CMR-tuned (test)	7	0.377	0.495	0.448	–

Note: CV values average the saved predictions over ten seeds. Unit means and medians give each serial number equal weight. Test models are tuned, whereas CV models have fixed configurations; their difference does not isolate a population effect.

cycles and SN 065 two in two. Together they account for five of seven false alarms despite contributing only six of 114 normal cycles. Cycle weighting therefore conceals substantial variation between units.

Sampling seven units with replacement from the 27 training units containing N01, using their seed-averaged LightGBM rates, gave a mean unit FA0 at most 0.10 in eight of 100,000 draws (0.00008; Monte Carlo standard error 0.000028). The same count was obtained using cycle weighting. This describes the empirical training population; it is not a significance test of the tuned test result, whose unit mean is 0.264. The protocol, complete unit and destination-class tables, sensitivity analysis without replacement, and input/output hashes are retained in the E2 artefacts.

S1.8. Semisynthetic Coverage of Cluster Procedures

We fixed protocol E3 before simulation and compared six procedures: paired percentile, pairs bootstrap-t, wild-t with Rademacher and Webb weights, sign-swap inversion, and cluster BCa. Each of 40 scenarios used 5,000 replicates and 2,000 internal resamples. The profiles preserve the observed exposure of each SN: the historical training set has 54 units, 14 with critical and 28 with normal cycles; the audited population has 65 units, 16 with critical and 35 with normal cycles. Historical folds are retained as metadata; inference pools external predictions rather than testing folds separately.

Beta distributions fitted by moments to seed-averaged LightGBM loss rates model between-unit heterogeneity. Gaussian copulas pair arm probabilities with latent correlation 0.8, with 0.2 as a sensitivity analysis. Affine probability transformations plant exact marginal differences of 0, ±0.01, ±0.02, and ±0.04 in CMR, and 0, ±0.01, ±0.03, and ±0.06 in FA0. Paired Bernoulli losses

retain ten dependent seeds through a mixture of shared and seed-specific cycle losses. The estimand is the cycle-weighted difference, not the unit-average difference.

Studentisation uses the cluster influence of the ratio of totals; wild-t resamples unrestricted ratio residuals and recentres each pseudo-sample. These adaptations are not the null-restricted regression bootstrap in Cameron et al. (2008). Sign-swap exactness additionally requires symmetry under the tested null, which the beta generator does not impose. BCa acceleration uses leave-one-SN-out ratios. Undefined intervals produce abstention, with unbounded limits, $p = 1$ and no equivalence, and are counted.

Eligibility required coverage of at least 0.93, false equivalence at the analytical margin of at most 0.05, Holm familywise error of at most 0.05, and abstention of at most 0.01 in every applicable scenario. Equivalence families have sizes 1, 3, 4 and 8; Holm families have 3, 4 and 8 rate contrasts with a shared reference under the global null. These homogeneous families do not reproduce the manuscript's mixed families containing DR and macro-F1. Wilson intervals quantify Monte Carlo uncertainty.

Table S7: Worst-case calibration across the semisynthetic scenarios

Procedure	Min. coverage	Max. boundary Eq.	Max. Holm FWER	Max. abstention
Percentile	0.850	0.014	0.099	0.000
Pairs bootstrap-t	0.873	0.008	0.044	0.035
Wild-t Rademacher	0.864	0.015	0.069	0.000
Wild-t Webb	0.863	0.016	0.066	0.000
Sign swap	0.881	0.011	0.049	0.000
BCa	0.814	0.011	0.099	0.001

Note: Minima and maxima over the applicable scenarios, including both exposure profiles and both copula correlations. Boundary Eq.: false equivalence when the true effect equals a margin. FWER: probability of any Holm rejection under the global null. Abstention is included in coverage; it is subject to a separate limit.

No procedure met all criteria. For historical CMR at zero effect and latent correlation 0.8, percentile coverage was 0.8942, against 0.9708 for wild-t Rademacher. At a true difference of -0.04, their coverages fell to 0.8502 and 0.8728, respectively: good null calibration did not ensure coverage away from zero (Figure S2). The percentile procedure's Holm familywise error reached 0.0992 in the historical FA0 profile with latent correlation 0.2 and eight contrasts (Wilson 95% interval 0.0912–0.1078), against a nominal 0.05. Consequently, no replacement procedure was selected, and the historical bootstrap table remains a descriptive calculation rather than

validated confirmatory inference.

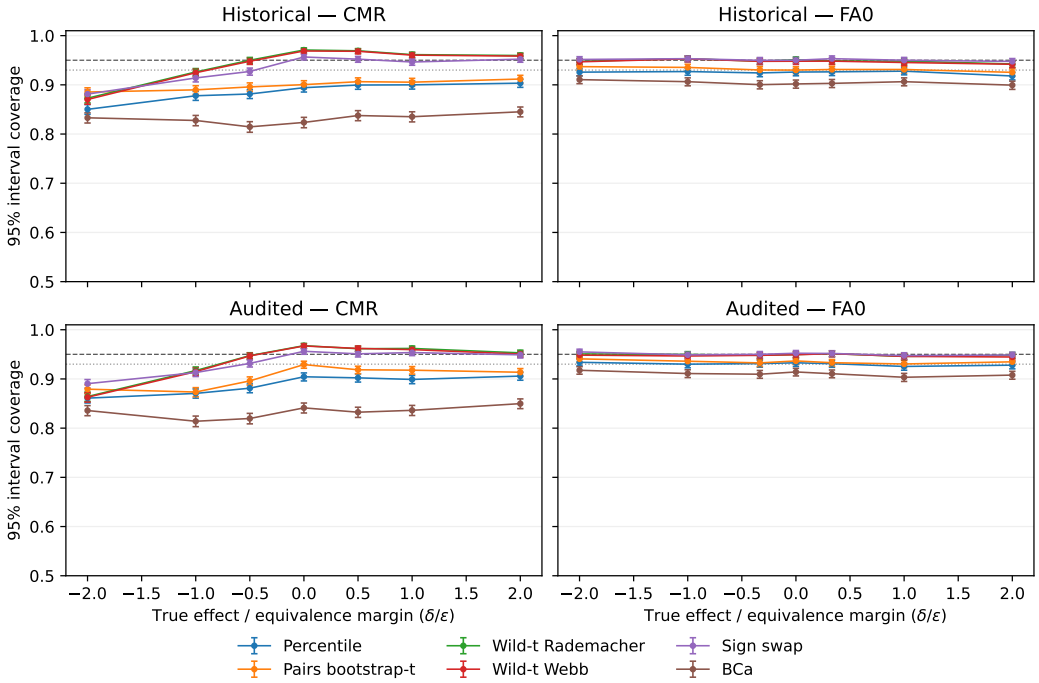


Figure S2: 95% interval coverage in the primary scenarios (latent correlation 0.8). Bars give Wilson 95% Monte Carlo intervals. Horizontal lines mark nominal coverage 0.95 and the eligibility minimum 0.93.

The generator fixes exposures and models loss heterogeneity independently of those exposures; it does not reproduce their observed association, the complete training process, or dependencies between distinct physical units. Fitting beta parameters to observed rates does not remove estimation noise. These findings therefore diagnose the tested procedures under stated scenarios, without establishing their performance for every plausible data-generating process. DR and macro-F1 were not validated. Full results, probabilities by replicate, protocol and hashes are retained in the E3 artefacts.

S1.9. Independent-Unit Evidence Requirements

Two different quantities determine the evidence requirements in Section 5.11. For independent evaluation units with a common probability p_{unit} of at least one observed miss, zero events in n units give the two-sided 95% Clopper–Pearson upper bound $U(n) = 1-0.025^{1/n}$. To obtain $U(n) \leq p_0$,

the minimum is $n_{\text{eval}} = \lceil \ln(0.025)/\ln(1-p_0) \rceil$. This is conditional on zero events, not a power calculation or an assurance that no events will occur.

For CRC with zero calibration loss, the corrected criterion has floor $1/(n+1)$, so $n_{\text{cal}} = \lceil 1/\alpha - 1 \rceil$ is necessary for an admissible threshold. The loss is the fraction of critical cycles missed within a unit; the guarantee concerns its expectation for a new exchangeable unit. It is neither a confidence bound for a binary event probability nor an assurance of useful false-alarm performance. The alarm-all fallback remains possible below this minimum.

Table S8: Independent-unit requirements for two distinct guarantees

Target	n_{eval}	Deficit vs 16	n_{cal}	Deficit vs 16
0.30	11	0	3	0
0.20	17	1	4	0
0.10	36	20	9	0
0.05	72	56	19	3
0.02	183	167	49	33
0.01	368	352	99	83

Note: Target denotes p_0 for evaluation and α for CRC, not a common estimand. Evaluation requires zero units with an observed miss; calibration assumes zero average unit loss. Deficits compare each role separately with the 16 critical units available (12 F03, four F06), as an idealised budget, not a feasible allocation of the same units to all roles. Training, calibration and evaluation units must be disjoint. Targets are individual, not simultaneous confidence bounds.

For the two critical test units, $U(2) = 0.8419$. Hypothetically, zero events in all 16 critical units would give $U(16) = 0.2059$; separately, zero events in 12 F03 units or four F06 units would give 0.2646 or 0.6024. These calculations do not assert that these groups had zero misses. Their unequal exposures, including 73 and one critical cycle in the test, prevent interpreting the binary bound as cycle-weighted CMR or risk at a standardised future exposure. If allocated solely to calibration, 16 units would give a best-case criterion floor of $1/17 = 0.0588$; the proposed calibration counts 7, 9 and 11 give floors 0.125, 0.100 and 0.0833.

For planning comparisons, Table S3 retains the historical $\delta_{\min} = \max(|\text{lo}|, |\text{hi}|)$: the margin required to contain the interval. With the strict interval criterion, the analytical margin must exceed this value. It is not a minimum detectable effect or a sample-size calculation. For CMR, RF–LightGBM and RF–XGBoost have historical values 0.1081 and 0.0806, against the analytical margin 0.02. F246–F115 gives 0.0265 for LightGBM

and 0.0246 for RF. Because the coverage assessment selected no calibrated CMR/FA0 procedure and did not assess DR or macro-F1, extrapolating sample size from these intervals would not yield defensible confirmatory requirements. Such planning requires a validated procedure, an effect of interest, power, exposure and heterogeneity assumptions, and an independent allocation of units to each role. The A1 artefacts retain all 15 primary historical contrasts and the analytic checks.

S1.10. Historical Exploratory Risk Calculation

The historical leave-one-unit-out calculation with 15 calibration units (Table S9) selected the alarm-all threshold at $\alpha = 0.0625$. At $\alpha = 0.10$, mean FA0 was 0.318 against 0.306 for the base model, and the mean unit loss was 0.076 against 0.082. These fold-specific scores describe that calculation; they do not establish the theorem's guarantee.

Table S9: Historical exploratory unit-level calculation

α	**Realised mean unit loss**	**Units with loss $> \alpha$ (of 16)**	**Mean FA0**
0.0625	0.000	0.0	1.000
0.10	0.076	2.3	0.318
0.15	0.079	1.3	0.314
0.20	0.082	1.3	0.306
0.30	0.082	0.7	0.306
No control (untuned, no threshold)	0.082	—	0.306

Note: Historical exploratory calculation with fold-specific untuned LightGBM scores, 16 severity-3 units and three seeds; leave-one-unit-out calibration uses 15 units. Realised loss is the mean share of missed severity-3 cycles in the held-out unit; the number exceeding α is averaged over seeds. Applicability of the CRC guarantee to these scores has not been established.

S1.11. Repeated Fixed-Model Risk Calibration

The E6 protocol repeats uniform partitions of the 65 audited units, including 16 critical units, with 200 repetitions at each calibration count. For each repetition, a permutation of the critical units assigns two to evaluation, $n_{\text{cal}} \in \{7, 9, 11\}$ to calibration, and the remaining seven, five or three to training. A separate permutation assigns ten of the 49 non-critical units to evaluation and 39 to training. The same permutations pair evaluation units across calibration counts. All cycles of a serial number share its role; there is

no class stratification or rejection of partitions lacking a training class. These partitions reuse the same observed population and do not add independent units.

Each model uses the historical untuned LightGBM configuration, frequency–severity weights, training-only Gaussian augmentation of F04/F05/F06 to 160 cycles per available class, and training-only median imputation. Missing training classes are not synthesised. The model is fitted once per partition and remains fixed during calibration and evaluation. The candidate thresholds are infinity, distinct critical calibration scores in descending order, and zero. The largest threshold satisfying the corrected criterion is selected; an empty admissible set invokes the zero-loss alarm-all fallback. An admissible selection of zero is recorded separately from an empty set.

Observed mean unit loss was below the target in all twelve calibrated conditions. At seven calibration units and $\alpha = 0.125$, mean unit loss was 0.005244, compared with 0.173744 for the base models, while mean FA0 increased from 0.325308 to 0.749995; the fifth and 95th partition percentiles of FA0 were 0.404518 and 0.993066. At nine and eleven calibration units with $\alpha = 0.10$, mean unit losses were 0.002685 and 0.027239 and mean FA0 was 0.811800 and 0.494290, respectively. Only the seven-unit condition below the criterion floor selected threshold zero; it did so in all 200 partitions. All other selected thresholds were positive or infinite. Normal-cycle denominators were non-zero throughout. These observations describe the available population; they do not establish expected-loss control or useful alarm burden for future units.

Training lacked F06 in 12, 30 and 77 of the 200 partitions at calibration counts seven, nine and eleven, respectively. At eleven, training also lacked F03 in three partitions. N02 was absent in 30, 34 and 38 partitions, respectively, and F04 in one partition at each count. No partitions were excluded or redrawn. The changing training coverage prevents attributing differences solely to calibration size; the full class-availability counts accompany the model manifests.

The partition assignments, complete calibration curves, cycle-level scores, per-unit losses, class availability and model hashes are retained in the E6 artefacts. The physical-unit budget and the simultaneous change in critical training size restrict interpretation; exchangeability for future F03/F06 units and operational utility require independent evidence.

Table S10: Repeated fixed-model CRC: means over 200 partitions per calibration count

n_{cal}	α	Unit loss	CMR	FA0	Alarm-all
7	0.100	0.000	0.000	1.000	1.000
7	0.125	0.005	0.008	0.750	0.000
7	0.150	0.045	0.049	0.411	0.000
7	0.200	0.106	0.098	0.345	0.000
7	Base	0.174	0.148	0.325	0.000
9	0.100	0.003	0.004	0.812	0.000
9	0.125	0.035	0.034	0.435	0.000
9	0.150	0.068	0.064	0.375	0.000
9	0.200	0.114	0.105	0.349	0.000
9	Base	0.182	0.162	0.329	0.000
11	0.100	0.027	0.026	0.494	0.000
11	0.125	0.055	0.054	0.400	0.000
11	0.150	0.081	0.076	0.374	0.000
11	0.200	0.119	0.111	0.351	0.000
11	Base	0.198	0.181	0.331	0.000

Note: Unit loss averages the shares of missed critical cycles in the two evaluation critical units. CMR and FA0 are cycle-weighted within each partition and then averaged across partitions. Base denotes no threshold promotion. Alarm-all is the fraction selecting threshold zero, including any empty-set fallback; at $n_{\text{cal}} = 7$, $\alpha = 0.10$ is below the 0.125 criterion floor. Training has seven, five or three critical units, respectively, so contrasts between calibration counts also change training composition. There are no confidence intervals or confirmatory tests based on reused physical units.

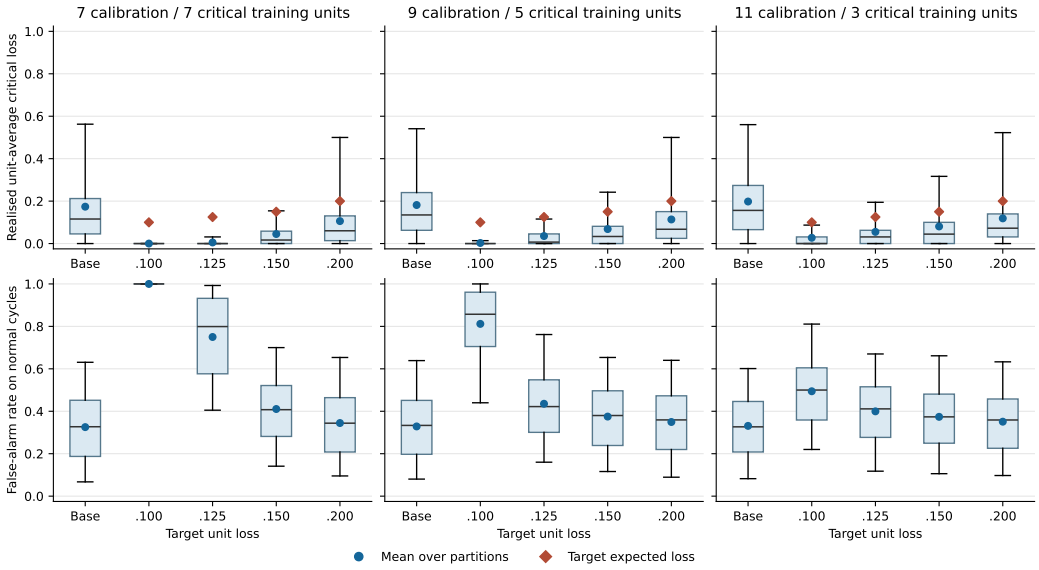


Figure S3: Distributions across the repeated E6 partitions. Boxes show the interquartile range, horizontal bars the median, and whiskers the fifth and 95th percentiles; points show means and diamonds the target expected unit loss. Outlying partition values are omitted from the display but retained in the data. These are descriptive distributions, not confidence intervals. The CRC guarantee concerns expected unit loss, not the fraction of partitions or units whose realised loss is below the target.

S1.12. Temporal Feature Inventory

Each PBIT cycle is a time-ordered sequence of $T \approx 123$ scalar measurements. The signal is then converted into a fixed-length feature vector with 246 numeric signal-derived descriptors (the set F246) in eleven families:

- whole-cycle statistics (12): mean, standard deviation, extrema, quantiles, root-mean-square value, energy, and trapezoidal area;
- endpoint and reversal geometry (10): start and end values, deltas, peak and valley values, and their positions;
- eight direction-aware segment summaries of 12 descriptors each (96), computed on eight contiguous segments of the cycle;
- segment contrast (17) between the outbound and return strokes and between mirrored segments;
- first- and second-difference dynamics (7 and 4): average and extreme slopes, acceleration descriptors, and sign-change rates;
- rolling-window summaries over windows of 5 and 15 samples (35 each);
- local-shape descriptors (14): peak and valley counts and prominences;
- linear-trend descriptors (6): slope, intercept, coefficient of determination, and residual spread;
- frequency descriptors (5): dominant bin of the fast Fourier transform, spectral centroid, bandwidth, entropy, and high-frequency energy ratio;
- five indicators of missing readings, which are constant in the retained data because missing readings are interpolated beforehand.

S1.13. F06 Signal Archetypes and Auxiliary Checks

The checks below support the interpretation of the leave-one-unit-out results in Section 5.10 by describing signal coverage and class separability.

The dataset contains nine F06 records from four serial numbers, of which seven are retained after the quality audit (Section 3.4). SN 045 is the flat archetype: experiment 917 near −3.66 V, experiment 920 at zero output (retained; results without it are reported where they differ), and experiments 950–951 near 0.64 V. SN 052 (experiment 1178), SN 054 (experiment 1227), and SN 056 (experiment 1258) share a wide-range oscillatory archetype, with a signal range of 2.75–3.27 V, and contribute one cycle each.

When a single F06 unit is left in training, none of the 12 source–target pairs yields an F06 prediction for the target, including pairs of oscillatory

units. A binary F03-versus-F06 task gives an F06 recall of 1.0 for SN 052, 0.6 to 1.0 for SN 054, 0.0 for SN 045 and 0.0 to 0.2 for SN 056, so F06 is separable from F03 for two of the four units and not for the others. Holding SN 045 out also removes most of the N02 cycles (59 of the 66 retained): all 59 are misclassified, so any inference about this unit mixes the loss of the flat F06 archetype with the loss of N02.

S1.14. Policies Selected on Inner Data: Design Details and All Budgets

Design.. The population is the 1,195 audited cycles of the 54 historical training units (experiments 931, 949 and 1107 excluded); the 11 historical test units are not used. Folds are the five grouped outer folds of the ten seeds (42, 0, 7, 13, 21, 1, 2, 3, 4, 5) with five grouped inner folds. In each training set, F04, F05 and F06 are augmented to 160 cycles per class present with 1% Gaussian noise and descriptors are median-imputed from training data only. The base classifier is LightGBM with 400 trees, learning rate 0.05, 63 leaves, at least five samples per leaf, row and column subsampling 0.8 (row subsampling active) and severity–frequency weights. Four arms are tuned by the tree-structured Parzen estimator (100 trials, no pruning) over the space of Section S1.1, with objective macro-F1 or CMR and severity-3 weight fixed at one or tuned in $[1, 20]$; the F05 weight is tuned in $[1, 10]$ in all four. A trial must keep the inner balanced accuracy no more than 0.037 below that of the untuned model. The score for thresholding is $P(\text{F03}) + P(\text{F06})$ with promotion to the more probable severity-3 class; the threshold minimises inner CMR, then FA0, subject to the inner FA0 budget. The cost arms use the cost matrix of Section 4.7; MetaCost uses ten bootstrap models of 200 trees. Inner-budget infeasibility keeps the unthresholded prediction of the policy's classifier and is counted. Search budgets differ between the tuned arms (100 trials) and the cost arms (five cost ratios), so no equal-budget claim is made. The 50 outer fits (104,651 model fits, 56.9 hours) were verified against recorded hashes and by regenerating model probabilities before any outer metric was computed. The independent recomputation of CMR, $\bar{L}$, FA0, per-unit FA0 and downward risk by scalar loops for six policy–budget combinations in all seeds (60 pairs) agreed to 2×10^{-16}.

Descriptive analysis.. Metrics are computed per seed on the predictions pooled over the five outer folds and averaged over seeds. Seeds are not independent replicates. Intervals resample units (10,000 paired draws, seed 20261027) and are reported as uncalibrated. Three semisynthetic examinations of the paired

CMR comparison found none calibrated: six procedures in Section S1.8; four restricted wild-bootstrap and jackknife variants (minimum coverage 0.855 to 0.879 for reductions of 0.04 with correlated arms); and four procedures for the unit-average loss $\bar{L}$ (minimum coverage 0.875 to 0.893 in the same scenarios). The second and third examinations followed a known negative result, and the choices of estimand and of the fallback procedure for FA0 were made after it. No test, p-value, multiplicity adjustment, equivalence or superiority statement is derived for the policies.

Retrospective reference curve.. The curve of the untuned model, built with the outer labels, starts at its unthresholded operating point and can only add alarms. For each policy it gives the lowest CMR at an FA0 not above that of the policy. The CMR of the policy exceeds the curve by 0.027 (CMR-tuned, fixed severity-3 weight), 0.013 to 0.014 (CMR-tuned, tuned weight), 0.028 to 0.039 (macro-F1-tuned), 0.011 to 0.037 (minimum expected cost) and 0.006 to 0.038 (MetaCost) over the three budgets. The curve is undefined in a seed when the FA0 of the policy is below the unthresholded FA0 of the reference, which happened for one seed of MetaCost at 0.40 (post-result clarification, recorded in the run manifest). Because the curve uses outer labels, it favours the reference and is used only to locate the policies, never to declare an improvement.

S1.15. Descriptive Reproduction of the SMOTEWB–LightGBM Pipeline

A post-hoc descriptive reproduction of the pipeline of Li et al. (2024) (SMOTE with boosting (SMOTEWB) oversampling with LightGBM, using the `smote-variants` reference implementation and 100 TPE trials per outer fold; ten seeds) gives, under unit-grouped folds, macro-F1 0.362 and CMR 0.097 for the retuned pipeline, against 0.380 and 0.084 for the Gaussian-augmentation reference of Section 5.3, with FA0 higher by 0.069 (cluster-resampling interval 0.024 to 0.121; descriptive, not calibrated). Under random folds the same pipeline raises macro-F1 by 0.386 and lowers FA0 from 0.530 to 0.137, while the CMR change (-0.028; -0.060 to $+0.008$) is not resolved: the leakage pattern of Section 5.2 recurs in the published method. Under grouped folds, between 59% and 93% of the cycles generated by SMOTEWB for the minority classes are exact copies of real cycles. These checks did not use the data or tuning protocol of the original study. Table S12 consolidates the reproduced pipeline results and differences.

Table S11: Policies selected on inner data, all FA0 budgets

Budget	Policy	Fallback /50 folds	FA0 > budget /10 seeds	CMR	$\bar{L}$	FA0	ΔCMR	$\Delta\bar{L}$	ΔFA0
0.40	untuned	33	9	0.080	0.133	0.463	—	—	—
0.40	macro_F1_fixed	35	10	0.062	0.064	0.532	−0.018 [−0.042; −0.003]	−0.069 [−0.201; −0.002]	0.069 [0.023; 0.132]
0.40	macro_F1_optimized	37	10	0.058	0.084	0.545	−0.022 [−0.050; −0.002]	−0.048 [−0.134; 0.005]	0.083 [0.033; 0.138]
0.40	CMR_fixed	50	10	0.028	0.045	0.822	−0.052 [−0.107; −0.006]	−0.088 [−0.206; −0.011]	0.360 [0.251; 0.471]
0.40	CMR_optimized	50	10	0.015	0.026	0.817	−0.065 [−0.118; −0.019]	−0.107 [−0.243; −0.017]	0.354 [0.246; 0.466]
0.40	minimum_cost	40	10	0.085	0.136	0.464	0.004 [0.001; 0.009]	0.003 [0.000; 0.007]	0.002 [−0.003; 0.007]
0.40	metacost	43	8	0.086	0.138	0.455	0.005 [0.001; 0.012]	0.005 [0.001; 0.011]	−0.007 [−0.015; −0.001]
0.30	untuned	47	10	0.086	0.138	0.452	—	—	—
0.30	macro_F1_fixed	46	10	0.063	0.064	0.521	−0.023 [−0.049; −0.004]	−0.074 [−0.206; −0.005]	0.069 [0.024; 0.133]
0.30	macro_F1_optimized	47	10	0.060	0.087	0.533	−0.025 [−0.056; −0.002]	−0.051 [−0.137; 0.004]	0.081 [0.034; 0.135]
0.30	CMR_fixed	50	10	0.028	0.045	0.822	−0.058 [−0.118; −0.007]	−0.093 [−0.211; −0.015]	0.370 [0.266; 0.479]
0.30	CMR_optimized	50	10	0.015	0.026	0.817	−0.070 [−0.128; −0.021]	−0.112 [−0.248; −0.020]	0.365 [0.260; 0.474]
0.30	minimum_cost	47	10	0.085	0.136	0.453	−0.001 [−0.002; 0.000]	−0.002 [−0.006; 0.000]	0.001 [0.000; 0.002]
0.30	metacost	50	10	0.086	0.138	0.451	0.000 [0.000; 0.000]	0.000 [0.000; 0.000]	−0.001 [−0.003; 0.000]
0.50	untuned	3	5	0.047	0.072	0.501	—	—	—
0.50	macro_F1_fixed	13	8	0.045	0.043	0.567	−0.002 [−0.014; 0.008]	−0.029 [−0.096; 0.006]	0.066 [0.018; 0.129]
0.50	macro_F1_optimized	11	8	0.049	0.069	0.575	0.002 [−0.005; 0.009]	−0.002 [−0.023; 0.016]	0.075 [0.026; 0.129]
0.50	CMR_fixed	47	10	0.028	0.045	0.823	−0.019 [−0.056; 0.011]	−0.026 [−0.066; 0.006]	0.322 [0.216; 0.433]
0.50	CMR_optimized	47	10	0.014	0.025	0.820	−0.033 [−0.069; −0.006]	−0.046 [−0.102; −0.003]	0.319 [0.214; 0.429]
0.50	minimum_cost	13	7	0.068	0.110	0.511	0.021 [0.007; 0.038]	0.039 [0.007; 0.088]	0.010 [−0.014; 0.035]
0.50	metacost	14	6	0.076	0.097	0.496	0.029 [0.009; 0.053]	0.025 [0.008; 0.045]	−0.005 [−0.033; 0.025]

Note: Means over ten seeds; fallback folds are of 50 outer folds; "FA0 > budget" counts seeds (of ten) in which the external FA0 exceeded the budget. Intervals are paired unit-resampling percentiles without validated coverage. Every fold is retained, including fallbacks; no common feasible frontier or operational ranking is inferred.

Table S12: E8 pipeline reproduction on the present PBIT population

Pipeline	Partition	CMR	FA0	Macro-F1	Micro-F1
Gaussian reference	Grouped by SN	0.084	0.460	0.380	0.542
Li fixed	Grouped by SN	0.076	0.519	0.371	0.515
Li retuned	Grouped by SN	0.097	0.530	0.362	0.491
Li fixed	Random	0.067	0.117	0.761	0.847
Li retuned	Random	0.070	0.137	0.748	0.830

Note: Source: E8 summary and frozen protocol. Historical training population: 1,196 cycles/54 SN, ten seeds and five outer folds; results are means of metrics pooled within seeds. Grouped and random designs each use corresponding inner folds. Retuning uses 100 TPE trials per outer fold and micro-F1 selection. No original-study performance is reproduced: the data and partition differ, Min–Max scaling uses training data only, and the search budget is specified locally. SMOTEWB uses the `smote-variants` reference implementation; the Gaussian reference uses severity–frequency weights. This is a descriptive pipeline comparison, not a claim of original-study replication or superiority.

S1.16. Complete Triage Policies: Budget and Recall Sensitivity

E10 uses 1,195 cycles/54 SN and the ten paired grouped outer/inner splits of E1. Each seed pools 139 F01, 259 F02, 194 F03, 29 F04, 30 F05 and six F06 cycles. F06 is present in three units. All eight arms failed joint admissibility internally in all 50 folds and externally in all ten seeds at every recall/budget combination. The identical values across recall targets arise from the predefined infeasible fallback, not three successful recall-controlled policies.

Table S13: E10 observed fallbacks at all three alarm budgets

Policy	Budget	FA0	Min. recall	F02	F05	F06 detect	F06 exact
Historical control	0.30	0.452	0.373	0.373	0.637	0.833	0.133
Matched promotion	0.30	0.453	0.376	0.376	0.637	0.850	0.133
Complete	0.30	0.343	0.208	0.208	0.507	0.817	0.133
Tuned complete, CMR fixed	0.30	0.317	0.103	0.103	0.490	0.683	0.150
Tuned complete, CMR optimised	0.30	0.318	0.075	0.075	0.607	0.600	0.150
Tuned complete, F1 fixed	0.30	0.357	0.202	0.202	0.470	0.800	0.133
Tuned complete, F1 optimised	0.30	0.379	0.201	0.201	0.493	0.783	0.150
Joint selection	0.30	0.373	0.218	0.218	0.543	0.750	0.117
Historical control	0.40	0.463	0.379	0.379	0.657	0.850	0.167
Matched promotion	0.40	0.470	0.400	0.400	0.637	0.850	0.133
Complete	0.40	0.436	0.354	0.354	0.583	0.833	0.133
Tuned complete, CMR fixed	0.40	0.413	0.188	0.188	0.607	0.733	0.150
Tuned complete, CMR optimised	0.40	0.414	0.151	0.151	0.643	0.733	0.150
Tuned complete, F1 fixed	0.40	0.449	0.311	0.311	0.610	0.883	0.133
Tuned complete, F1 optimised	0.40	0.472	0.303	0.303	0.607	0.817	0.150
Joint selection	0.40	0.490	0.369	0.369	0.720	0.867	0.167
Historical control	0.50	0.501	0.399	0.399	0.760	0.933	0.267
Matched promotion	0.50	0.532	0.478	0.478	0.683	0.883	0.150
Complete	0.50	0.531	0.476	0.476	0.680	0.883	0.150
Tuned complete, CMR fixed	0.50	0.520	0.303	0.303	0.640	0.817	0.167
Tuned complete, CMR optimised	0.50	0.514	0.260	0.260	0.713	0.817	0.167
Tuned complete, F1 fixed	0.50	0.544	0.428	0.428	0.677	0.950	0.133
Tuned complete, F1 optimised	0.50	0.558	0.410	0.410	0.667	0.867	0.150
Joint selection	0.50	0.543	0.417	0.417	0.740	0.900	0.133

Note: Means over ten seeds of metrics pooled across outer folds. Values are identical for recall targets 0.85, 0.90 and 0.95 because none was admissible. Minimum recall denotes the mean of each seed's minimum detection recall over F01–F05. F06 detection and exact-class recall are separate exploratory endpoints. No folds or seeds were excluded. The source CSVs retain all nine scenarios, per-class normal alarm rates, critical losses and per-unit results.

The frozen batch completed 25,000 internal fits and 137 selected-model outer refits. A separate five-fit instrumentation probe followed a Windows memory-reporting error after the ten-trial timing pilot; it did not change selection or inspect outer performance. All fits and the prior frozen version

were preserved. Independent scalar recomputation of the class-detection rates, exact recalls, FA0, CMR and severity-3 recall across 720 seed–arm–scenario combinations agreed within 2.23×10^{-16}; 10,574 recorded output hashes matched. These checks establish computational consistency, not external validity or confidence-interval coverage. No additional trials or class exclusions were made in response to the failed requirements.

S1.17. Exploratory Error and Signal Diagnosis after E10

E12 examined F02/F05 in the same 1,195-cycle development population after their E10 results were known. No independent test cohort or new classifier was used. For each of 50 grouped outer partitions, references were real training cycles from other SN. The raw signals contained 123 ordered measurement positions and no missing values in this frozen population; positions are not independently documented acquisition times. Feature scaling used median imputation and IQRs of the real training set only. Normal-reference quantiles gave equal total weight to each training normal unit and excluded same-SN neighbours.

Table S14: E12 geometric diagnostics of focal cycles against real training references

Class	Representation	Normal nearer	Normal $q_{.95}$	Nearer: missed	Nearer: detected
F02	Raw level	0.393	0.903	0.493	0.210
F02	Within-cycle shape	0.720	0.955	0.728	0.704
F02	F18 / training IQR	0.564	0.966	0.628	0.448
F02	F246 / training IQR	0.592	0.977	0.683	0.426
F05	Raw level	0.800	0.733	0.704	0.869
F05	Within-cycle shape	0.600	1.000	0.784	0.469
F05	F18 / training IQR	0.970	1.000	0.976	0.966
F05	F246 / training IQR	0.983	1.000	1.000	0.971

Note: Proportions of cycle–seed comparisons; repeated seeds are dependent. "Normal nearer" includes ties against the nearest same-fault training reference. "Normal $q_{.95}$" is the share whose normal distance does not exceed the unit-weighted 95th percentile of training cross-SN normal-to-normal distances. It is a descriptive scale reference, not an acceptance region or guarantee. The last two columns condition normal-nearer proportions on complete-policy miss/detection. Absolute distances across differently scaled representations are not comparable.

The two F02 units with supplied normal records were SN 011 and SN 020, covering 40 of the 259 F02 cycles. All three F05 units had normal records. Within-unit comparisons are oracle diagnostics, not references available to the unit-held-out training procedure, and no verified transition chronology was inferred. Illustrative raw/shape pairs were fixed by an error/proximity rule:

IDs 173 and 183 for F02, and 1466 and 1437 for F05, with neighbours from the corresponding seed-42 training fold. Their selection is not representative sampling; plots, IDs, neighbours and all unit-level counts are retained with the diagnostic outputs.

Independent verification checked 413 input hashes and 17 output hashes, recomputed 120 arm–class–unit error groups with maximum numerical difference 7.11×10^{-15}, and reproduced 160 sampled nearest-reference pairs within 6.67×10^{-16}. ID/SN/class fields agreed for all 1,195 records across the supplied original CSV, XLSX, clean long table and features. No focal cross-label matches were found at exact precision or after rounding to six or three decimals. These are checks of computation and export consistency; the exports may share the same label origin and are not independent physical confirmations.

S1.18. Global Quality and Numerical-Region Controls

E13 audited raw readings from the 54 development SN, including cycles 931, 949 and 1107, without admitting the 11 historical test units. Rules used only readings and indices: any missing value; whole-cycle standard deviation below 10^{-6}; at least ten consecutive values differing by no more than 10^{-9}; isolated spikes; and sequence equality at exact or six-decimal precision. A spike required opposite signs in adjacent differences, both larger than eight times 1.4826 MAD of the cycle's finite first differences; this rule was undefined when its scale did not exceed 10^{-12}. These are numerical indicators, not instrument tolerances. Saturation could not be confirmed without instrument limits, and repeated-cycle changes were not treated as acquisition errors without documented comparable conditions.

The 1,198 raw cycles contained 248 unusable entries in 11 cycles. Values and missingness agreed with the supplied original CSV within 10^{-12}. The clean table had filled nine isolated entries and excluded the two nearly incomplete cycles; no other finite reading changed beyond that tolerance. E13 identified 65 processable suspects and two cycles with more than half their readings absent. The single observed value in cycle 931 gives a numerical standard deviation of zero but does not establish whole-cycle constancy. Complete constant cycles 920 and 1107 were both suspect irrespective of class. No equality groups or structural index errors were found, and no entire unit was excluded.

E14 retained the 1,195-cycle historical development population. Its uniform control used indices 1–41, 42–82 and 83–123. The empirical control

learned two common boundaries separately in each real training set by minimising

$$\sum_{i\in\mathcal{T}} a_i \sum_{k=1}^{3} \sum_{t\in I_k(b_1,b_2)} \left(z_{it} - \bar{z}_{ik}\right)^2, \qquad a_i = \frac{1}{m\, n_{u(i)}},$$

where m is the number of real training SN, $n_{u(i)}$ counts their cycles, and $\bar{z}_{ik}$ is the regional mean. Each z_i is centred and scaled within its cycle; standard deviations below 10^{-6} yield a zero alignment vector. All admissible boundary pairs with region lengths of at least 15 were enumerated, with lexicographic tie-breaking. Neither labels nor synthetic or external signals entered this objective. Predictive descriptors used the original levels.

Each region supplied its mean, median, standard deviation, quartiles, extrema, range, RMS, slope per index, last-minus-first reading, mean and maximum absolute first difference, and fraction of adjacent differences below 10^{-9}. Differences of regional means and standard deviations for the three region pairs gave 48 attributes in total. The uniform and empirical controls had identical dimensionality; their combination with F246 had 294 attributes. Across 300 training partitions, the first boundary ranged from 15 to 46 (median 16), and the second from 55 to 87 (median 59); 129 first boundaries were at the minimum length. They were not interpreted as verified physical transitions.

Table S15: E14 numerical-region controls at recall target 0.90 and FA0 budget 0.40

Configuration	FA0	F01	F02	F03	F04	F05	F06 (expl.)
F246 (M0)	0.436	0.977	0.354	0.905	0.969	0.583	0.833
Uniform regions (M1)	0.443	0.993	0.364	0.846	1.000	0.403	0.833
Empirical regions (M2)	0.448	0.983	0.389	0.855	1.000	0.450	0.833
F246 + empirical (M3)	0.440	0.982	0.373	0.898	0.986	0.577	0.833

Note: Cycle-pooled detection recalls and FA0, averaged over the same ten seeds; repeated seeds are dependent. Population: 1,195 cycles/54 SN. F06 is exploratory. All four arms had zero internally admissible outer partitions out of 50 and zero externally compliant seed evaluations out of ten in all nine scenarios. F246 reuses the verified complete-policy scores; 900 new fits concern the other three arms.

E15 restored cycle 1107 as suspect, producing 1,196 processable candidate cycles of the same units; 931/949 remained analytically unusable by the class-independent coverage rule. Q0 retained all candidate training cycles,

and Q1 removed the 65 suspects before augmentation. Both policies kept every candidate evaluation cycle, including suspects, internally and externally. Cycle 1107 inherited the SN 045 roles from the historical assignments. F246 and empirical descriptors used the same expanded signals within each policy, with imputers fitted only on that training set. Thus Q1–Q0 measures the complete filtering procedure on a common evaluation population, including its changed training composition and learned limits; it does not identify removal of instrumental noise.

Table S16: E15 training-quality sensitivity on a common 1,196-cycle evaluation population

Configuration	FA0	F01	F02	F03	F04	F05	F06 (expl.)
Unfiltered F246 (Q0)	0.439	0.982	0.354	0.904	0.966	0.600	0.833
Filtered F246 (Q1)	0.440	0.978	0.332	0.905	0.969	0.587	0.933
Unfiltered empirical (Q0)	0.447	0.988	0.395	0.853	1.000	0.457	0.833
Filtered empirical (Q1)	0.432	0.989	0.364	0.866	1.000	0.450	0.733

Note: Primary 0.90/0.40 scenario; detection recalls and FA0 averaged across ten dependent seeds. Suspects are removed only from Q1 training. Both policies evaluate all 1,196 processable cycles, so comparison with E14 additionally changes the population. No whole-SN exclusion rule was justified or applied. Each arm had zero admissible internal partitions out of 50 and zero compliant external seed evaluations out of ten in every scenario. E15 used 1,200 new fits.

E16 returned to the 1,195-cycle E14 population and its fixed training-derived boundaries. A single-region model retained 14 attributes; omission of a region removed all its descriptors and dependent contrasts, leaving 30. Artificially shifting both boundaries by -1 or $+1$ retained 48 attributes and non-empty regions, even when one region became 14 positions long. The original training boundaries and full-cycle augmentation remained shared components: these are attribute ablations conditional on that procedure, not experiments acquiring or training from only one physical stroke. They establish no early-decision protocol.

The three comparisons used 4,500 new scientific fits, including their pilots; two additional fits were timing controls for E15 and did not evaluate performance. Independent computation checks reconstructed all 7,650 internal scenario selections, verified 1,828,710 saved external decisions, and checked detection/false-alarm metrics in 1,530 seed-level rows, with maximum scalar discrepancy 1.11×10^{-16}. These are checks of supplied data and computed outputs, not independent physical label confirmation, model guarantees or

Table S17: E16 numerical-region ablations at the primary 0.90/0.40 setting

Configuration	FA0	F01	F02	F03	F04	F05	F06 (expl.)
All regions (reference)	0.448	0.983	0.389	0.855	1.000	0.450	0.833
R1 only	0.387	0.960	0.385	0.539	0.903	0.410	0.700
R2 only	0.387	0.974	0.375	0.711	0.993	0.390	0.833
R3 only	0.433	0.970	0.390	0.812	1.000	0.547	0.567
Without R1	0.445	0.982	0.392	0.828	1.000	0.447	0.817
Without R2	0.427	0.986	0.403	0.851	1.000	0.480	0.833
Without R3	0.399	0.996	0.349	0.766	0.997	0.423	0.833
Both limits −1	0.440	0.992	0.394	0.863	1.000	0.440	0.833
Both limits +1	0.443	0.983	0.391	0.848	1.000	0.427	0.833

Note: Population: 1,195 cycles/54 SN; means over the same ten dependent seeds. The all-region reference reuses verified E14 M2 scores. The eight other arms used 2,400 new fits with their own inner-selected thresholds. No arm was internally admissible or externally compliant in any of the nine scenarios. Changes in recall must be read with changes in FA0; neither variable selection nor artificial boundary shifts identify physical phases or causes.

external validation.

S1.19. Rerun of the Main Comparisons on the Audited Population

The historical comparisons of the main text use 1,196 training cycles, including experiment 1107 (an N02 cycle of SN 045 historically treated as a dead channel), which the quality audit excludes. The leakage, classifier-family, feature-set and imbalance experiments were rerun on the 1,195 audited cycles of the same 54 units, with the ten seeds, models, fixed hyperparameters, weights and augmentation unchanged; only the experiment was removed. With one cycle fewer, `StratifiedGroupKFold` can assign units to folds differently for the same seed, so differences reflect partition variability as well as the removed cycle. The decision rule was fixed before the runs: each audited estimate had to lie within the historical range of the ten per-seed values, and each designated contrast had to keep its sign and the position of its descriptive interval relative to zero and to the analytical margins.

All 259 estimates (28 for leakage, 14 for classifier families, 49 for feature sets, 168 for imbalance treatment) lie within the historical per-seed range; the largest absolute difference is 0.012 (severity-3 recall of unweighted LightGBM). Of the 33 designated contrasts, 29 keep sign and interval position (all 10 for leakage, all 4 for classifier families, all 3 for imbalance treatment). Four of the 16 feature-set contrasts change, all for LightGBM and all near zero

(Table S18): F246 minus F115 moves from −0.002 to +0.001 in CMR and from +0.002 to −0.003 in downward risk (a change of sign within the margin), and from straddling zero to lying within the margin in FA0 (−0.010 to 0.000); F246 minus F18 in macro-F1 moves from +0.008 (within the margin) to +0.011 (interval −0.007 to +0.030, straddling zero). None establishes a population ranking: the F246–F115 point contrasts remain small and the LightGBM F18 comparison remains descriptive. The tables of the main text therefore keep the historical population. Intervals are paired unit-resampling percentiles, descriptive and not calibrated (Section S1.8).

Table S18: Designated contrasts, historical and audited populations

Block	Contrast	Metric	Historical [interval]	Audited [interval]	Same
leakage	LightGBM: random - grouped	Accuracy	0.298 [0.213; 0.386]	0.302 [0.217; 0.390]	yes
leakage	LightGBM: random - grouped	macro_F1	0.376 [0.186; 0.457]	0.375 [0.180; 0.462]	yes
leakage	LightGBM: random - grouped	CMR	−0.023 [−0.069; 0.010]	−0.025 [−0.073; 0.009]	yes
leakage	LightGBM: random - grouped	FA0	−0.341 [−0.462; −0.217]	−0.348 [−0.468; −0.224]	yes
leakage	LightGBM: random - grouped	DR	−0.098 [−0.153; −0.050]	−0.100 [−0.155; −0.052]	yes
leakage	Random Forest: random - grouped	Accuracy	0.313 [0.234; 0.393]	0.310 [0.231; 0.391]	yes
leakage	Random Forest: random - grouped	macro_F1	0.392 [0.211; 0.478]	0.390 [0.209; 0.475]	yes
leakage	Random Forest: random - grouped	CMR	−0.022 [−0.067; 0.012]	−0.021 [−0.062; 0.010]	yes
leakage	Random Forest: random - grouped	FA0	−0.336 [−0.445; −0.217]	−0.332 [−0.441; −0.216]	yes
leakage	Random Forest: random - grouped	DR	−0.101 [−0.149; −0.060]	−0.101 [−0.150; −0.060]	yes
families	Random Forest - LightGBM	CMR	−0.010 [−0.096; 0.060]	−0.017 [−0.108; 0.054]	yes
families	Random Forest - LightGBM	FA0	0.042 [−0.011; 0.102]	0.040 [−0.009; 0.097]	yes
families	Random Forest - XGBoost	CMR	−0.004 [−0.072; 0.062]	−0.004 [−0.076; 0.059]	yes
families	Random Forest - XGBoost	FA0	−0.022 [−0.064; 0.024]	−0.013 [−0.054; 0.032]	yes
ablation	LightGBM: F246 - F115	CMR	−0.002 [−0.019; 0.009]	0.001 [−0.012; 0.013]	no
ablation	LightGBM: F246 - F115	FA0	−0.010 [−0.030; 0.010]	0.000 [−0.021; 0.022]	no
ablation	LightGBM: F246 - F115	DR	0.002 [−0.010; 0.013]	−0.003 [−0.013; 0.006]	no
ablation	LightGBM: F246 - F115	macro_F1	0.013 [−0.003; 0.033]	0.017 [−0.001; 0.042]	yes
ablation	LightGBM: F246 - F18	CMR	−0.049 [−0.100; 0.004]	−0.048 [−0.097; 0.008]	yes
ablation	LightGBM: F246 - F18	FA0	−0.036 [−0.089; 0.016]	−0.032 [−0.085; 0.021]	yes
ablation	LightGBM: F246 - F18	DR	−0.003 [−0.026; 0.023]	−0.002 [−0.025; 0.022]	yes
ablation	LightGBM: F246 - F18	macro_F1	0.008 [−0.012; 0.029]	0.011 [−0.007; 0.030]	no
ablation	Random Forest: F246 - F115	CMR	−0.002 [−0.019; 0.012]	−0.002 [−0.014; 0.008]	yes
ablation	Random Forest: F246 - F115	FA0	0.001 [−0.014; 0.016]	0.004 [−0.012; 0.021]	yes
ablation	Random Forest: F246 - F115	DR	0.004 [−0.004; 0.011]	0.002 [−0.005; 0.011]	yes
ablation	Random Forest: F246 - F115	macro_F1	−0.003 [−0.010; 0.002]	−0.003 [−0.009; 0.002]	yes
ablation	Random Forest: F246 - F18	CMR	−0.080 [−0.124; −0.034]	−0.086 [−0.132; −0.037]	yes
ablation	Random Forest: F246 - F18	FA0	0.030 [−0.059; 0.133]	0.028 [−0.060; 0.131]	yes
ablation	Random Forest: F246 - F18	DR	0.010 [−0.023; 0.052]	0.010 [−0.023; 0.054]	yes
ablation	Random Forest: F246 - F18	macro_F1	−0.007 [−0.027; 0.014]	−0.008 [−0.028; 0.014]	yes
imbalance	LightGBM: product - frequency weights (Gaussian)	CMR	−0.002 [−0.015; 0.010]	−0.002 [−0.015; 0.010]	yes
imbalance	LightGBM: Gaussian - none (product weights)	macro_F1	0.005 [−0.009; 0.016]	0.004 [−0.009; 0.016]	yes
imbalance	LightGBM: Gaussian - SMOTE (product weights)	macro_F1	−0.002 [−0.022; 0.015]	−0.003 [−0.023; 0.016]	yes

Note: Differences are means over ten seeds; intervals are paired unit-resampling 95% percentiles (not calibrated). "Same" indicates that the sign and the position of the interval relative to zero and to the analytical margin (0.02 for CMR and downward risk, 0.03 otherwise) are unchanged.

S1.20. Fixed-Model Leakage Contrast

The random and grouped arms of Section 5.2 train different models: the random arm sees every unit, the grouped arm roughly 80% of them. The difference in accuracy could therefore reflect the number of units seen in

training as well as leakage. Following Vieira et al. (2026), who keep the trained model fixed and change only the test set, the contrast was repeated with a single model. In each of the five grouped folds of the ten seeds (historical population of 1,196 cycles), the cycles of the training units were split, stratified by fault mode, into 80% for fitting and 20% reserved. A model was trained on the fitting cycles only (augmentation, weights and fixed hyperparameters as in the main comparison) and evaluated without refitting on the reserved cycles of the training units (seen units) and on all cycles of the held-out units (unseen units). No reserved cycle belonged to a unit absent from the fitting set, and the held-out units were disjoint from the fitting units in every fold.

The accuracy contrast was fixed as the primary endpoint before the runs: the fraction of the random-minus-grouped gap of the main text reproduced by the seen-minus-unseen contrast of the same model, read as at least 0.8 (the gap is leakage), 0.4 to 0.8 (partly diversity) or below 0.4 (mostly training composition). The fractions are 1.06 for LightGBM and 1.12 for Random Forest in accuracy, and between 1.01 and 1.42 for the other metrics (the largest, for the CMR of LightGBM, concerns a difference whose interval includes zero) (Table S19). The main text therefore keeps the magnitude of the leakage gap. Seen and unseen sets differ in class composition, which affects accuracy and macro-F1; balanced accuracy over the eight classes present in both sets gives differences of 0.364 (LightGBM) and 0.428 (Random Forest), so the gap is not an artefact of class mix. The CMR differences (-0.033 and -0.025) have the sign of the main text and intervals that include zero. Intervals are paired unit-resampling percentiles, descriptive and not calibrated (Section S1.8). An independent recomputation of accuracy, CMR and FA0 by scalar loops for both models and both sets agreed to 10^{-17}.

S1.21. Fully Nested Leave-One-Unit-Out Analysis of F06

Each of the four units with F06 was held out in turn, and the models were fitted on the cycles of the other 64 units of the audited population (1,487 cycles, 65 units), with five seeds (42, 0, 7, 13, 21). The hyperparameter search ran inside the 64 units: tree-structured Parzen estimator, 100 trials per arm, five inner grouped folds stratified by fault mode, augmentation of F04, F05 and F06 to 160 cycles per class inside each inner training set, median imputation and severity–frequency weights. Two arms were searched as in the main text: macro-F1 as objective with the severity-3 weight fixed at one, and CMR as objective with that weight tuned in $[1, 20]$; the F05 weight was tuned

Table S19: Fixed-model leakage contrast

Model	Metric	Seen units	Unseen units	Difference [interval]	Manuscript gap	Fraction
LightGBM	Accuracy	0.841	0.526	0.315 [0.235; 0.399]	0.298	1.06
LightGBM	macro_F1	0.752	0.365	0.387 [0.184; 0.464]	0.376	1.03
LightGBM	CMR	0.067	0.100	−0.033 [−0.085; 0.010]	−0.023	1.42
LightGBM	FA0	0.114	0.459	−0.345 [−0.465; −0.225]	−0.341	1.01
LightGBM	DR	0.073	0.185	−0.112 [−0.169; −0.063]	−0.098	1.15
LightGBM	SE	0.150	0.454	−0.304 [−0.381; −0.227]	−0.289	1.05
LightGBM	R3	0.862	0.766	0.096 [0.010; 0.191]	0.078	1.24
LightGBM	BalancedAcc	0.739	0.375	0.364 [0.143; 0.450]	—	—
Random Forest	Accuracy	0.832	0.481	0.351 [0.273; 0.430]	0.313	1.12
Random Forest	macro_F1	0.779	0.359	0.420 [0.236; 0.499]	0.392	1.07
Random Forest	CMR	0.058	0.083	−0.025 [−0.084; 0.021]	−0.022	1.10
Random Forest	FA0	0.145	0.502	−0.357 [−0.470; −0.242]	−0.336	1.06
Random Forest	DR	0.078	0.205	−0.128 [−0.185; −0.076]	−0.101	1.26
Random Forest	SE	0.158	0.494	−0.336 [−0.413; −0.261]	−0.300	1.12
Random Forest	R3	0.867	0.714	0.153 [0.054; 0.273]	0.150	1.01
Random Forest	BalancedAcc	0.797	0.369	0.428 [0.267; 0.507]	—	—

Note: Ten seeds, 54 units. Seen units: reserved cycles of the training units; unseen units: all cycles of the held-out units; the same model is evaluated on both. The difference is seen minus unseen. Manuscript gap: random minus grouped of Table 5. Fraction: difference divided by manuscript gap. Intervals are paired unit-resampling percentiles and are not calibrated.

in $[1, 10]$ in both. A trial had to keep the inner balanced accuracy within 0.037 of that of the untuned model. The selected model was refitted on all 64 units and predicted the held-out unit by the most probable class, without a threshold. All 20 runs completed, each with 100 completed trials per arm, no failed search and a held-out unit absent from every training, augmentation and selection step (asserted in the code). The runs took between 1.5 and 3 hours each (five in parallel). Three units with F06 remained in the training set of every run.

The selected severity-3 weight was 1 in the F1-tuned arm by construction and had median 15.1 (range 7.0 to 19.9) in the CMR-tuned arm, close to the upper bound of its search interval; the median number of eligible trials was 96 in that arm. The interpretation rule was fixed beforehand: severity 3 predicted in at least four of five seeds for each oscillatory unit in both arms means that the result survives nested tuning. It was met: the seed counts are 5, 5 and 4 (F1-tuned) and 5, 5 and 5 (CMR-tuned) for SN 052, SN 054 and SN 056. Intervals and tests were not computed: five seeds per unit are not independent replicates and there are four units.

S1.22. Sensitivity to the Set of Critical Classes

The severity mapping of the main text is the authors' proposal. Protocol E17, fixed before any computation, recounts the saved predictions and probabilities under four alternative sets of critical classes, with normal classes unchanged: M1 {F03, F05, F06}, M2 {F02, F03, F06}, M3 {F02, F03, F04, F05, F06} and M4 {F06}; M0 {F03, F06} is the reference, and its recount reproduced the published values. No model was refitted and no threshold or cost ratio was reselected, so the models keep the weights and severity-3 boost defined under M0. FA0, accuracy, macro-F1, the triage comparison and the diagnosis of F02/F05 do not depend on the mapping. Intervals are descriptive (10,000 paired resamples of serial numbers). Under M4 the seven F06 cycles make the contrasts uninformative; recalibrating CRC on the fixed E6 models with 1.8 to 2.8 F06 calibration units left only the alarm-all fallback in 94.5% to 100% of the partitions. M1–M3 cannot be recalibrated on those partitions, whose calibration units were restricted to F03 and F06.

Table S20 classifies each conclusion of the main text. The design-choice conclusions and the reading of CMR with FA0 hold under M1–M3, with two isolated reversals in feature contrasts. With F02 critical, leakage also lowers the apparent CMR, the held-out result without critical misses no longer holds, and the critical units exceed the CRC minimum of 19.

S1.23. Data-Quality Audit

A data-quality audit was applied to every cycle before modelling. Cycles with more than 50% of their readings missing were excluded (experiments 931 and 949, both F06 from SN 045, with 122 and 117 missing readings). A cycle whose signal is constant (historically treated as a dead channel) was excluded when it belongs to a normal class: experiment 1107, an N02 cycle from SN 045 fixed at 12.94 V. One further constant-signal cycle, an F06 cycle from SN 045 (experiment 920, all-zero output), is retained because zero output is physically consistent with command-signal disconnection; analyses that exclude it are reported where they differ. The nine isolated missing readings in nine retained cycles are linearly interpolated before feature extraction. The retained dataset has 1,487 cycles, of which seven are F06. The subsequent label-independent sensitivity in Section 4.10 re-examines this class-dependent constancy rule without changing the historical population. Four retained cycles (experiments 20, 994, 997 and 1258) show an abrupt drop of the signal to approximately zero for at least ten consecutive samples; they occur in three classes (F01, N01 and F06) and three units and are kept.

Table S20: Sensitivity of the conclusions to the set of critical classes

Conclusion under M0	M1 {F03, F05, F06}	M2 {F02, F03, F06}	M3 {F02–F06}
RQ1: no established CMR direction under leakage (RF −0.022 [−0.066; +0.012])	Unchanged (−0.043 [−0.109; +0.008])	Reversed: −0.193 [−0.292; −0.103]	Reversed: −0.181 [−0.273; −0.098]
Families within uncertainty (RF − LightGBM −0.010)	Unchanged	Unchanged	Unchanged
F246 − F115 unresolved (−0.002)	Unchanged	Reversed: +0.019 [+0.001; +0.039]	Unchanged
F246 − F18 unresolved (−0.049)	Reversed: −0.054 [−0.102; −0.005]	Unchanged	Unchanged
LightGBM below baseline B4 (−0.091)	Unresolved (−0.083 [−0.193; +0.044])	Unchanged, larger (−0.177)	Unchanged, larger (−0.166)
Severity weights do not lower CMR (−0.002)	Unchanged	Unchanged (−0.030)	Unchanged
CMR tuning lowers CMR only with higher FA0 (0.030 vs 0.070)	Unchanged (0.048 vs 0.117)	Unchanged (0.196 vs 0.349)	Unchanged (0.183 vs 0.340)
Inner-selected CMR-tuned policies lower CMR (0.052; 0.065)	Unchanged (0.059; 0.077)	Unchanged, larger (0.205; 0.197)	Unchanged (0.188; 0.184)
Held-out test: no critical miss of the CMR-tuned model, 2 units (bound 0.84)	Unchanged (3 units; bound 0.71)	Reversed: 18 cycles missed in 3 of 5 units	Reversed: 18 cycles missed in 3 of 7 units
Nested LOSO preserves criticality of oscillatory F06 units	Unchanged	Unchanged	Unchanged
Fewer than 19 critical units (16; 14 in training)	Reversed in the population (20), unchanged in training (17)	Reversed (30; 25 in training)	Reversed (37; 30 in training)

Note: CMR differences with descriptive 95% intervals from resampling serial numbers; models and predictions fixed. "Reversed" marks a change of direction or of qualification relative to M0, as fixed in protocol E17. M4 {F06} is not shown because its contrasts are uninformative.

S1.24. Methodological Details Moved from the Main Text

The following subsections give, verbatim, the methodological details summarised in the Methodology section of the main text.

S1.24.1. Stratification by Severity

Stratifying by the four-level severity index instead changes the CMR by +0.015 (descriptive 95% interval −0.048 to +0.093) for Random Forest and −0.014 (−0.089 to +0.056) for LightGBM (ten paired seeds), so the choice of label does not alter the conclusions.

S1.24.2. Resampling and Calibration of the Inferential Procedures

A retrospective correction reuses the saved predictions with 10,000 paired cluster resamples (seed 20261010), preserving the same unit weights across all ten fixed seeds before averaging their metrics. Differences between two configurations are reported as point differences with the 2.5th and 97.5th percentiles of the paired resampled differences (95% intervals).

These intervals are descriptive: three semisynthetic assessments (Supplementary Material, Sections S1.8 and S1.14) found that none of the procedures examined (six, four and four in the three assessments, including restricted wild bootstraps, jackknife variants and Student-t tests of the unit-mean loss) met the prespecified calibration criteria for paired differences in CMR, with minimum coverage between 0.81 and 0.89 for reductions of the loss when the two arms are strongly correlated. The restricted wild bootstrap was calibrated for FA0 in every scenario of the second assessment, but its confirmation on independent seeds was not run, because the protocol required a procedure eligible for CMR. Coverage for DR and macro-F1 was not assessed. The paper therefore states no equivalence, superiority, non-inferiority or significance and applies no multiplicity adjustment. The margins of ±0.02 for CMR and downward risk and ±0.03 for macro-F1 and FA0 serve only as reading thresholds, informed by the seed variability of the selected model (S1.3); they are not operator-approved limits. A historical analysis of equivalence (a Bonferroni-adjusted interval criterion for two one-sided tests, Lakens, 2017) and of difference (centred bootstrap tests with Holm adjustment, Holm, 1979), computed during the retrospective correction, is kept in the Supplementary Material (Section S1.5) for traceability and is not used for claims. No resample lacked an endpoint denominator.

S1.24.3. Augmentation and Compared Imbalance Treatments

Gaussian-noise augmentation is applied to the three rarest minority classes, F04, F05, and F06, inside the training fold only. A synthetic cycle is generated from a real training cycle with valid samples $x_1, \ldots, x_T$ and within-cycle standard deviation s_x by perturbing the signal,

$$\tilde{x}_t = x_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}\Big(0,\ (\rho_{\mathrm{n}}\, s_x)^2\Big), \quad \rho_{\mathrm{n}} = 0.01, \tag{S1}$$

Here $\tilde{x}_t$ is the synthetic reading at position t, and the independently drawn ε_t has zero mean and standard deviation equal to 1% of the source cycle's standard deviation s_x, computed with the number of valid readings as the variance divisor. An absolute standard deviation of ρ_{n} in the original signal units is used when $s_x \leq 10^{-9}$, and the feature vector is recomputed from $\tilde{x}$. Each of F04, F05, and F06 is expanded to 160 training cycles per fold with 1% noise, a level conservative enough to preserve the physical character of the signal while increasing intra-class diversity. Test folds remain fully real and unaugmented.

The treatment of imbalance is evaluated rather than assumed (Section 5.3). Four weighting schemes are compared: no weights; inverse class frequency only ($\alpha_\sigma \equiv 1$ in Eq. (9)); severity multiplier only ($n_{\mathrm{tr}}/(K\, n_{y_i})$ replaced by one); and the product of both, which is the default. Three augmentation schemes are compared: none; the signal-domain Gaussian noise of Eq. (S1); and SMOTE applied in feature space to the real training cycles of F04, F05, and F06 up to 160 cycles each, with $\min(5, n-1)$ neighbours, where n is the number of real training cycles in the minority class, and median imputation of the missing descriptors (Chawla et al., 2002).

Signal-domain noise was preferred to interpolation for two reasons. SMOTE interpolates linearly between a minority-class sample and one of its k nearest neighbours, and because each position of a PBIT cycle encodes a specific commanded or sensed quantity, interpolating between two fault-class sequences may produce a sequence corresponding to no physical failure mode, a limitation highlighted for time-series augmentation by Fawaz et al. (2018). Gaussian noise perturbs an observed sequence locally, is equivalent to Tikhonov regularisation (Bishop, 1995), and has been reported to outperform SMOTE and ADASYN in other domains (Becerra-Suarez et al., 2025). These arguments motivate the default but do not establish it for PBIT data; Section 5.3 tests it.

S1.24.4. Search Space and Constraint of the Nested Search

LightGBM hyperparameters are selected by a Bayesian search with the tree-structured Parzen estimator (TPE) of Optuna (Akiba et al., 2019). Two objectives are used: the mean macro-F1 over the folds (50 trials, the F1-tuned configuration) and the mean CMR (100 trials, the CMR-tuned configuration). For the CMR objective, a *severity-boost* multiplier λ_{sev3} is introduced into the per-sample weights as an additional factor on all severity-3 samples (F03 and F06), together with a boost λ_{f05} on F05. Both multiply the base severity–frequency weight:

$$w_i = \alpha_{\sigma(y_i)} \frac{n_{\text{tr}}}{K\, n_{y_i}} \lambda_{\text{sev3}}^{\mathbb{1}[\sigma(y_i)=3]} \lambda_{\text{f05}}^{\mathbb{1}[y_i=\text{F05}]}, \tag{S2}$$

where $\mathbb{1}[\cdot]$ is one when its condition holds and zero otherwise. The severity-3 factor therefore applies only to F03/F06, and the F05 factor only to F05; the normal classes keep a strictly positive weight. The F1-targeted search also tunes λ_{f05}, but fixes $\lambda_{\text{sev3}} = 1$. The historical comparison changes the objective, trial budget, severity-3 boost space and feasibility rule together. It therefore compares two search procedures and cannot attribute their difference to the objective alone. The search covers $\lambda_{\text{sev3}} \in [1, 20]$, $\lambda_{\text{f05}} \in [1, 10]$, the number of trees in $[200, 800]$, the learning rate in $[0.01, 0.2]$ (logarithmic), `num_leaves` in $[15, 127]$, `min_child_samples` in $[3, 50]$, `subsample` and `colsample_bytree` in $[0.5, 1.0]$, and `reg_alpha` and `reg_lambda` in $[10^{-8}, 10]$ (logarithmic). For the CMR objective, a constraint on the mean balanced accuracy excludes degenerate configurations that predict severity 3 for most cycles. Row subsampling was inactive because `subsample_freq` remained zero. Inner-fold metrics were averaged even when the folds contained different sets of classes; folds without critical cycles were omitted from the CMR objective. The historical objective therefore differs from the pooled evaluation estimand.

Selecting hyperparameters and evaluating on the same folds gives an optimistic estimate (Cawley and Talbot, 2010). The search is therefore nested: five outer folds, assigned by serial number, define the evaluation; within each outer training set, an independent search with five inner grouped folds selects the configuration, which is trained on the outer training set and applied to the outer test fold. In each outer fold, the selected configuration is compared with the untuned model and with the parameters of the non-nested selection run (S1.2). The original rule (mean balanced accuracy ≥ 0.50) left no feasible trial in three of the five outer folds, because the inner validation folds contain only 4 to 8 of the 8 classes. It was replaced by a rule relative

to the untuned model: the inner balanced accuracy of that model minus 0.037, which is the slack of the original rule on the full training partition, as disclosed in Section 4.12.

S1.24.5. Selection and Fallback Rules of the Triage Comparison

For each candidate, inner out-of-sample probabilities determine thresholds for the nine combinations of minimum detection recall (0.85, 0.90, 0.95) and FA0 budget (0.30, 0.40, 0.50). The primary combination is 0.90/0.40. Admissibility requires every F01–F05 detection recall to meet its target and FA0 not to exceed its budget. Among admissible choices, minimise FA0, then maximise the minimum and mean recalls; threshold ties prefer the larger threshold, and model ties the earlier trial. If no admissible choice exists, retain the choice under the alarm budget with the largest minimum recall, then smallest FA0 and largest mean recall, explicitly recording infeasibility. If the unilateral family cannot meet even the alarm budget, retain its unpromoted prediction and record that failure. An absent primary denominator is undefined, never an attained requirement. The TPE objective directs search to the primary combination; the same 100 candidates serve all nine combinations. No outer labels choose configurations, thresholds or the set of primary classes. The protocol and code were frozen before new fitting; the fixed timing pilot contributed its first ten trials to the 100 of its partition. Scores, configurations, failed requirements and predictions are retained for all 50 outer partitions.

S1.24.6. Design of the Diagnosis of the Limiting Classes

The triage results identified F02 and F05 as the limiting primary classes (Section 5.6). They were then selected for a descriptive investigation (protocol E12) of the saved predictions at the primary 0.90/0.40 setting, asking whether their misses are concentrated in a few units or spread across the class. All eight arms were decomposed by serial number; the complete policy and joint selection were fixed as reporting anchors before the diagnostics. Cycle-weighted detection recall was compared with an equal-unit average, and each unit's share of misses with its share of cycle exposure. Repeated misses across the ten seeds describe reuse of the same cycles, not independent evidence.

For each outer-fold focal cycle, nearest real training records from normality and from the same fault class were compared under four fixed representations: the ordered raw signal, within-cycle centred and scaled shape, F18, and F246. Distances were root-mean-square coordinate differences. Feature imputations

and Interquartile-range (IQR) scales used only real training records; zero-IQR coordinates were excluded. No target-unit record entered these reference sets. A normal reference was formed from training normal-to-normal distances excluding the same serial number, with equal total weight per unit when computing its quantiles. This was a geometric diagnostic, not a new classifier, acceptance region or test of physical equality. Within-unit normal comparisons, when available, were kept as retrospective oracle checks. Signal matching at exact precision and after rounding to six and three decimal places, and agreement of experiment identifier, SN and class across the supplied tables, examined internal traceability rather than independent label confirmation. No new model or decision threshold was fitted, and the triage requirements were retained.

S1.24.7. Design of the Quality and Numerical-Region Controls

Two explanations of the F02/F05 limitation remained open after the diagnosis: defective records and descriptors that miss regional structure within the test sequence. Four exploratory protocols (E13–E16 in the Supplementary Material) address them. In the label-independent signal audit (E13), we audited the raw records of the same 54 development units, including the three historically excluded cycles, using rules whose inputs contain indices and readings but no class labels or predictions. Missingness, constancy, repeated sequences and isolated spikes provided numerical flags rather than confirmed acquisition diagnoses. The supplied bench sequences cover extension and contraction, but no verified phase-index map was available. Legacy stroke descriptors therefore remain index-based proxies; new numerical regions were not labelled as physical strokes or reversal.

The regional-representation comparison (E14) compared F246 with 48 descriptors of three uniform regions, the same descriptors of three regions learned without labels from real training signals, and their combination with F246 (294 attributes). Each training set learned its own common limits, with at least 15 positions per region and equal total alignment weight per SN. Within-cycle normalisation served only to learn these limits; predictive attributes retained absolute levels. The regional ablations (E16) used each learned region alone, omitted each region together with its dependent contrasts, or shifted both limits by one position. These artificial perturbations were not physical tolerances.

The comparison and ablations retained the 1,195-cycle population. The training-quality sensitivity (E15) used 1,196 processable cycles by restoring

cycle 1107 as suspect, then crossed F246 and the empirical descriptors with complete or filtered training. The filter removed flagged real training cycles before augmentation; every arm retained all processable internal and external evaluation cycles. No entire SN was excluded. The added cycle inherited the grouped roles of its SN rather than generating new folds. All comparisons retained the complete-policy configuration, weights, augmentation and inner-selected triage decision, with the same recall requirements and alarm budgets. They were specified after earlier results were known and remain exploratory. Supplementary Material, Section S1.18, gives the regional objective, controls and population distinctions.

S1.24.8. Protocol of the Repeated Calibration Partitions

We additionally repeat unit partitions of the audited population 200 times for each calibration count $n \in \{7, 9, 11\}$, using the same evaluation units across counts. Each partition reserves two randomly selected critical units and ten non-critical units for evaluation, calibrates on n of the remaining critical units, and trains on all remaining units. All cycles of a serial number share one role. The untuned LightGBM configuration and preprocessing rules remain fixed; each model is fitted once and then held fixed during calibration and evaluation. Targets are $\alpha \in \{0.10, 0.125, 0.15, 0.20\}$, including targets below the criterion floor. The primary summary averages the realised unit-average critical loss across partitions; CMR, FA0 and alarm-all frequency describe the operating cost. Increasing calibration from seven to eleven units reduces critical training units from seven to three, so comparisons do not isolate calibration size. Repeated use of the same units provides a retrospective sensitivity analysis, not independent validation of the guarantee for future units.

S1.24.9. Complete List of Deviations from the Analysis Plan

This manuscript reports a retrospective analysis of retained code, predictions and result files. No retained dated record establishes that the historical margins, comparison families or interpretation rules preceded inspection of the results; the analyses were not registered externally. The corrected inference also uses previously inspected predictions and is not a prospective confirmation. The following historical decisions affect interpretation:

- The balanced-accuracy constraint of the hyperparameter search was replaced after the first nested run and after inspecting its aggregate

outer results (Section 4.6); the original-constraint run (nested CMR 0.055, with no valid search in three of five outer folds) is reported in the retained nested-validation summary.

- The false-alarm rate FA0 was introduced as a co-metric after the first nested results; its use in the nested and operating-point analyses follows from that observation and was not pre-specified at the outset.
- The training and test partitions include experiment 1107, which the quality audit excludes (Section 3.4); the comparisons were rerun on the audited population as a sensitivity analysis (Supplementary Material, Section S1.19).
- The leave-one-unit-out experiment of Section 5.10 was extended with the tuned models after the untuned models failed to reproduce the single-cycle success of the held-out test.
- Four analyses were added after an internal audit of the manuscript: the imbalance ablation (Section 5.3), the cost-sensitive comparators (Section 5.4), and the sensitivity of the conclusions to the equivalence margins (S1.5) and to the fixed hyperparameters (S1.6).
- Threshold references were selected using outer labels; the historical test was reused for sensitivity analyses and fixed-model CRC. Neither use supplies a new independently selected policy validation.
- The reproduction of the pipeline of Li et al. (2024) (Section 6.2) was added after the results above were known; its protocol was frozen before any test prediction, and its intervals are descriptive.
- The fully nested leave-one-unit-out analysis of F06 (Section 5.10) was added after the audit; its protocol, interpretation rule and hashes were frozen before the runs.
- The fixed-model leakage contrast (Supplementary Material, Section S1.20) was added after the audit and after the random-versus-grouped result was known; its protocol was frozen before any prediction was generated.
- The inference planned for the policies selected on inner data was replaced by a descriptive analysis after the coverage assessments validated no procedure; the amendment was frozen before any external metric was computed. One post-result clarification followed: the retrospective reference curve is undefined when a policy FA0 lies below the unthresholded FA0 of the reference, and averages use the defined seeds only (Supplementary Material, Section S1.14).
- Multiplicity-adjusted equivalence and centred-bootstrap difference tests

were introduced during the retrospective correction; once the coverage assessments validated no procedure, the main text was rewritten with descriptive differences and intervals only, and the historical tests were kept in the Supplementary Material.

- The fault-detection triage comparison was specified after the historical analyses. Its primary F01–F05 scope and exploratory F06 treatment, recall targets, alarm budgets, decision controls and fallback rules were fixed before its new fitting and outer evaluation. It addresses a new detection criterion, not a retrospective redefinition of historical CMR.
- The diagnosis of the limiting classes selected F02/F05 after their triage results were known. The subsequent error decomposition and four geometric representations were fixed before computing these diagnostics; all findings are exploratory. No labels, class scope, decision thresholds or performance targets were revised.
- The quality and numerical-region controls followed the earlier results. Numerical quality rules were fixed before the new audit, and the regional and filtering controls before their fitting. Their adaptation to empirical regions reflects the absence of a verified physical map. The training-quality sensitivity restored cycle 1107 in a separate sensitivity population; it did not redefine historical results. No unit was removed because of poor classification, and no new independent cohort or confirmatory inference was introduced.

Analyses from an earlier pipeline whose code and feature definitions were not retained are excluded (S1.26); F115 reconstructs an earlier feature set. The reported comparisons therefore concern the retained pipeline and the deviations above.

S1.25. Results Moved from the Main Text

The following subsections give, verbatim, the results summarised in the Results section of the main text.

S1.25.1. Design Choices: Complete Results

Overall Classification Performance. The classifier family is the first design choice examined for RQ2: ensemble and boosting methods have been applied to avionics fault classification (Section 2.2), and the comparison describes the CMR–FA0 pairs of their fixed configurations. Table 6 compares the four classifier families under the common protocol. The point estimates of CMR range from 0.071 (Random Forest and Gradient Boosting) to 0.081

(LightGBM), but the 95% intervals obtained by resampling serial numbers are about 0.13 wide and overlap almost entirely. The paired differences between Random Forest and LightGBM, and between Random Forest and XGBoost, are −0.010 (95% interval −0.098 to +0.059) and −0.005 (−0.072 to +0.060) in CMR, and +0.042 (−0.011 to +0.098) and −0.022 (−0.064 to +0.023) in FA0. All four intervals include zero, and Random Forest has the lower CMR in a share of 0.57 of the resamples of serial numbers against each gradient-boosting family. Random Forest has the lowest CMR in 8 of the 10 seeds, but seeds are not independent replicates: the intervals are about ten times wider than the differences between the family estimates (at most 0.010). The intervals are descriptive (Section 4.2), so neither superiority nor equivalence between families is established. The family comparison is conditional on the fixed settings: the minimum leaf size alone moves the CMR of Random Forest between 0.041 and 0.112 (S1.6), a range wider than the largest difference between the family estimates (0.010). The family contrasts therefore do not establish that one family is preferable once each is tuned. Per-class recall in a single seed is given in S1.4.

Feature-Set Ablation and Trivial Baselines. Feature richness is the second design choice of RQ2. F115 and F18 are reductions of F246: F115 removes the segment and short rolling-window families, and F18 keeps only 18 whole-signal statistics, so they test whether the added families are needed. The trivial baselines give the reference against which any classifier is read. The observed CMR contrast between F246 and F115 is small; the larger reductions relative to F18 are reported below as descriptive contrasts. Table 7 compares LightGBM and Random Forest on the 246 descriptors, the 115-descriptor reduction (F115) and 18 simple statistics (F18), together with five trivial baselines: always N01 (B1); a draw from the class priors (B2); a dispersion threshold that predicts N01 when the signal standard deviation is at most 1.5 times the median of N01 in the training fold and the most frequent fault class otherwise (B3); the same threshold with F03 as the alarm class (B4); and always F03 (B5).

For F246 minus F115 in LightGBM, the paired CMR difference is −0.002 (95% interval −0.019 to +0.010). The intervals of the differences in downward risk, FA0 and macro-F1 of both models are given in the Supplementary Material (Section S1.5). The comparison with F18 favours the larger sets in direction but is inconclusive for LightGBM: the CMR is lower by 0.049 for F246 (95% interval −0.101 to +0.003) and by 0.047 for F115. For Random

Forest, the difference between F246 and F18 in CMR is −0.080 (95% interval −0.124 to −0.033), an exploratory comparison. With 11 units of F03 and 3 of F06 in the training partition, the data cannot resolve differences of this size reliably.

The trivial baselines show why CMR must be read together with FA0. Always predicting F03 gives CMR = 0 with FA0 = 1. The threshold rule B4, which uses only the dispersion of the signal, reaches CMR = 0.172 (descriptive 95% interval 0.091 to 0.246) with FA0 = 0.388 and $R_3 = 0.828$, a severity-3 recall higher than that of the LightGBM models (0.708 to 0.801). LightGBM with 246 descriptors lowers the CMR by 0.091 (descriptive 95% interval −0.171 to −0.006) relative to B4, at a higher FA0 (0.461 against 0.388); its advantage lies in the identification of individual classes (macro-F1 0.378 against 0.127), not in the severity level. Between 46% and 50% of the normal cycles are predicted as faulty by all models and feature sets, a rate that is high for operational use and is invisible in accuracy or CMR.

Treatment of Imbalance: Sample Weights and Augmentation. The severity weights and the Gaussian augmentation were adopted as defaults on the arguments of Section 4.4; this subsection tests those arguments, which is the third design choice of RQ2. For LightGBM, the designated weight and augmentation contrasts have small point estimates; they do not establish a joint CMR–FA0 advantage. Table S21 reports the twelve combinations of sample weights and augmentation for LightGBM and Random Forest. For LightGBM, product weights against frequency weights, with Gaussian augmentation, differ in CMR by −0.002 (95% interval −0.015 to +0.010); Gaussian augmentation against none, with product weights, differs in macro-F1 by +0.005 (−0.008 to +0.016); and Gaussian augmentation against SMOTE differs in macro-F1 by −0.002 (−0.022 to +0.014). The intervals lie within the reading thresholds of Section 4.2, but no statement of equivalence is made. The CMR finding concerns the weight contrast and the two augmentation findings concern macro-F1; none speaks to joint CMR–FA0 behaviour. Severity weights are not supported as a means of lowering the critical miss rate beyond frequency weights. They raise FA0 by 0.032 (95% interval 0.020 to 0.044) relative to frequency weights and by 0.045 (0.027 to 0.066) relative to no weights, that is, the observed increase in alarms accompanies a CMR reduction of only 0.002 relative to frequency weights; so small a reduction does not rule out a CMR effect.

For Random Forest, product weights lower the CMR relative to frequency

weights by 0.015 (95% interval −0.024 to −0.006, exploratory) at the price of a higher FA0 (+0.059), and augmentation matters only when the weights are weak. The macro-F1 intervals between SMOTE and Gaussian noise for both models, and the CMR interval for Random Forest, lie within the reading thresholds. For LightGBM, the CMR difference is inconclusive (+0.017; 95% interval −0.009 to +0.060). These data do not confirm the argument against interpolation of Section 4.4: the LightGBM macro-F1 point differences are small, and their descriptive intervals permit neither an equivalence nor a superiority statement. These conclusions refer to one dataset and default hyperparameters, and the F06 recall, estimated from six cycles, cannot discriminate between the arms.

S1.25.2. Triage Comparison: Additional Results

This does not identify whether the limiting cause is feature overlap, condition-label validity, unit heterogeneity or the chosen model/search space. The same threshold choices were returned at recall targets 0.85, 0.90 and 0.95 for a given arm and alarm budget, because none was admissible and the predefined fallback does not use the unattained target. This equality is not evidence of robustness to a successfully controlled recall requirement. Full budget sensitivity is given in Table S13.

F06 detection recall ranged from 0.733 to 0.883 in the primary scenario, but exact F06 recall was only 0.133–0.167. These means reuse six cycles from three units and establish neither reliable learning of the specific class nor performance on future F06 units. The new comparison remains descriptive and does not replace the historical critical-miss or unit-calibration analyses.

S1.25.3. Diagnosis of the Limiting Classes: Proximity and Traceability

Geometric proximity was representation-dependent (Supplementary Material, Table S14). For F02, a normal training record was nearer than a same-class record, or tied, in 0.393 of raw-level comparisons, 0.720 after within-cycle shape normalisation, and 0.592 under training-scaled F246. For F05, the F246 fraction was 0.983; it was 1.000 among complete-policy misses but also 0.971 among detected cases. Proximity therefore does not itself explain the classifier's decisions or prove that the conditions are physically indistinguishable.

No discrepancies in experiment identifier, SN or class were found across the 1,195 supplied records, and no focal cross-label signal matches were found at the three examined precisions. These negative checks do not validate

Table S21: Treatment of imbalance

Weights	Aug.	CMR [95% CI]	FA0	DR	F1 [95% CI]	F04 recall	F06 recall
LightGBM							
None	None	0.092 [0.033; 0.167]	0.425	0.182	0.378 [0.287; 0.454]	0.324	0.150
None	Gaussian	0.091 [0.034; 0.168]	0.415	0.185	0.390 [0.295; 0.468]	0.393	0.167
None	SMOTE	0.080 [0.032; 0.138]	0.421	0.188	0.384 [0.288; 0.463]	0.448	0.183
Freq.	None	0.084 [0.036; 0.147]	0.443	0.183	0.377 [0.283; 0.457]	0.352	0.167
Freq.	Gaussian	0.086 [0.030; 0.162]	0.429	0.183	0.380 [0.287; 0.456]	0.345	0.150
Freq.	SMOTE	0.078 [0.037; 0.130]	0.430	0.188	0.382 [0.287; 0.462]	0.448	0.183
Sev.	None	0.085 [0.035; 0.152]	0.442	0.174	0.379 [0.286; 0.455]	0.317	0.150
Sev.	Gaussian	0.090 [0.028; 0.175]	0.451	0.180	0.383 [0.289; 0.458]	0.400	0.150
Sev.	SMOTE	0.085 [0.037; 0.139]	0.450	0.184	0.380 [0.285; 0.455]	0.438	0.167
Freq. × Sev.	None	0.071 [0.028; 0.129]	0.470	0.176	0.375 [0.282; 0.456]	0.345	0.167
Freq. × Sev.	Gaussian	0.084 [0.024; 0.166]	0.460	0.171	0.380 [0.289; 0.457]	0.324	0.167
Freq. × Sev.	SMOTE	0.067 [0.026; 0.115]	0.465	0.174	0.382 [0.287; 0.459]	0.448	0.167
Random Forest							
None	None	0.134 [0.042; 0.234]	0.398	0.201	0.349 [0.259; 0.414]	0.286	0.017
None	Gaussian	0.096 [0.032; 0.178]	0.419	0.210	0.372 [0.272; 0.455]	0.493	0.167
None	SMOTE	0.110 [0.037; 0.202]	0.423	0.214	0.371 [0.271; 0.450]	0.521	0.167
Freq.	None	0.091 [0.035; 0.170]	0.440	0.203	0.365 [0.268; 0.445]	0.410	0.167
Freq.	Gaussian	0.086 [0.028; 0.166]	0.444	0.209	0.365 [0.266; 0.447]	0.507	0.167
Freq.	SMOTE	0.096 [0.037; 0.180]	0.445	0.212	0.367 [0.266; 0.448]	0.534	0.167
Sev.	None	0.090 [0.027; 0.172]	0.446	0.196	0.339 [0.259; 0.394]	0.221	0.033
Sev.	Gaussian	0.082 [0.027; 0.163]	0.474	0.203	0.366 [0.268; 0.447]	0.510	0.167
Sev.	SMOTE	0.087 [0.033; 0.167]	0.481	0.206	0.367 [0.268; 0.449]	0.552	0.167
Freq. × Sev.	None	0.069 [0.021; 0.139]	0.492	0.187	0.379 [0.286; 0.444]	0.393	0.333
Freq. × Sev.	Gaussian	0.071 [0.015; 0.146]	0.503	0.199	0.359 [0.263; 0.439]	0.493	0.167
Freq. × Sev.	SMOTE	0.072 [0.017; 0.146]	0.505	0.203	0.362 [0.263; 0.443]	0.548	0.167

Note: 246 descriptors, ten paired seeds, 54 units and median imputation in all arms. Freq. and Sev. denote frequency and severity weights; Freq. × Sev. is their product. Aug.: augmentation; DR: downward risk; F1: macro-F1. F04/F06 recalls are pooled over evaluation folds; F06 has six cycles from three units. Brackets give 95% intervals from resampling serial numbers.

the physical labels. Only two of the 12 F02 units had supplied normal cycles in this development population, compared with all three F05 units; the records do not establish controlled before/after-fault comparisons. Acquisition settings, whether faults were real or induced, and confirmatory evidence remain undocumented. The findings support prioritising physical record review and unit/condition coverage; they do not justify relabelling records, removing difficult units or lowering the targets.

S1.25.4. Quality and Numerical-Region Controls: Complete Results

The raw development audit covered 1,198 cycles: 1,131 had no flag under the available rules, 65 were processable but suspect, and two lacked more than half their readings. The 248 unusable values in 11 cycles corresponded to `#VALUE!` entries in the original comma-separated values (CSV) file; that export evidence does not identify an instrumental cause. F02/F05 had neither missing readings nor wholly constant signals. SN 064 had one flagged cycle among 18, providing no quality-based basis for excluding the unit that contributed most F05 misses. The flags did not establish acquisition invalidity.

At the primary setting, empirical regions yielded F02/F05 detection recalls of 0.389/0.450 and FA0 of 0.448, compared with 0.364/0.403 and 0.443 for the equal-dimensional uniform control (Table S15). The combined 294-attribute representation gave 0.373/0.577 and 0.440, against 0.354/0.583 and 0.436 for F246. None met the joint requirements. Learning numerical limits did not validate a physical phase map: the first limit reached its minimum permitted length in 129 of 300 training partitions.

Filtering training alone did not resolve the difficult classes (Table S16). With F246, F02 recall fell from 0.354 to 0.332 and F05 from 0.600 to 0.587, while FA0 changed from 0.439 to 0.440. With empirical regions, filtering lowered FA0 from 0.447 to 0.432 but also lowered F02/F05 recall from 0.395/0.457 to 0.364/0.450. These within-filtering contrasts share all 1,196 evaluation cycles; comparisons with the regional representations additionally change the population.

The eight regional ablations produced mean F02 detection from 0.349 to 0.403 and FA0 from 0.387 to 0.445 (Table S17). Some reduced false alarms below 0.40 while leaving fault detection inadequate. No arm of these controls was internally admissible or met the external joint requirements in any of the nine scenarios. All 7,650 inner scenario selections were checked independently by explicit decisions at every distinct score threshold. These controls delimit the tested representations and filtering procedure; they neither prove physical

inseparability nor locate the faults within the actuator's motion.

S1.25.5. Held-Out Test: Unit Composition, F03 Confusion and Macro-F1

Unit composition also shapes FA0. The cycle-weighted FA0 of 0.461 for LightGBM under grouped validation falls to 0.410 without SN 045, which contributes 20.9% of its false alarms; this unit explains part of the inspection burden but does not account for the difference from the held-out test. For the F1-tuned test model, FA0 is 0.061 by cycle but 0.264 when units receive equal weight: SN 031 has three alarms in four normal cycles and SN 065 has two in two, so the seven test units are not uniformly easy (Supplementary Material, Section S1.7). Because grouped validation uses fixed models and the test uses tuned models, this contrast does not isolate a unit-population effect.

The severity level of the F03 cycles was preserved but their class was not: 61 of the 73 F03 cycles of SN 062 are predicted as F06 by both models (another severity-3 class), and the F03 class recall is 0 of 73 for the F1-tuned model and 6 of 73 for the CMR-tuned model. The severity-3 recall R_3 is 0.838 and 0.919, respectively. Training without augmentation of F06 gives the same confusion (61 to 63 of 73), so it is not an effect of the augmentation. Because all F03 cycles of the partition come from one unit, a property of the class cannot be separated from a signature of that unit.

The macro-F1 on the test partition (0.390 for the F1-tuned model) is close to the selection-fold value (0.427), but the correct classification of its single F06 cycle requires a further check of generalisation across units (Section 5.10).

S1.25.6. Non-Nested Leave-One-Unit-Out Analysis of F06

The following historical analysis is retained as a conditional diagnostic. To test whether F06 generalises to a unit not seen in training, each of the four F06 units was held out in turn and the models were trained on all cycles of the other 64 units (all classes), with five seeds for the augmentation noise and the models. The tuned arms reuse parameters selected on the historical training population, which included three of the four target units. These arms thus diagnose sensitivity to training composition conditional on those parameters; they do not provide a fully nested leave-one-unit-out estimate of the tuning procedure, which Table 14 reports above. Table S22 reports, for each held-out unit, the fraction of seeds in which its F06 cycles are predicted as F06 and as severity 3 (F03 or F06).

Table S22: Leave-one-unit-out prediction of F06

Held-out unit (archetype)	LightGBM untuned	RF	LightGBM F1-tuned	LightGBM CMR-tuned
SN 045 (flat, 4 cycles)	0.00 / 0.60	0.00 / 1.00	0.00 / 0.75	0.05 / 0.75
SN 052 (oscillatory)	0.00 / 0.00	0.20 / 1.00	0.00 / 1.00	0.20 / 1.00
SN 054 (oscillatory)	0.20 / 1.00	0.00 / 1.00	0.00 / 1.00	0.60 / 1.00
SN 056 (oscillatory)	0.00 / 0.00	0.00 / 1.00	0.00 / 1.00	0.00 / 1.00

Note: Cells report the fraction of five seeds, pooled over each unit's cycles, predicting F06 / severity 3. Training excludes the entire target unit. RF: Random Forest. Tuned LightGBM arms reuse selection-run parameters; three target units belonged to the population used for that selection, so these arms are conditional diagnostics rather than fully nested leave-one-unit-out evaluations; the fully nested counterpart is Table 14.

No F06 unit is recognised as F06 by the untuned LightGBM in more than one of five seeds (SN 054); SN 052 is predicted as F01 (severity 1) in all five seeds, whereas SN 056 is predicted as N01 (severity 0), giving critical misses in all five. The F1-tuned and CMR-tuned LightGBM models and the Random Forest predict severity 3 for the three oscillatory units in all seeds, almost always as F03, whereas the untuned LightGBM does not; some models thus preserve the severity level without recognising F06, and the experiment does not isolate which component is responsible. In these non-nested arms, the exact class F06 is predicted in at most three of five seeds (CMR-tuned model, SN 054).

In the leave-one-unit-out experiment, the hyperparameters that classified the SN 052 cycle correctly in the held-out test (54 training units; Section 5.9) predict F06 for it in none of five seeds (F1-tuned) and in one of five (CMR-tuned), and F03 otherwise. The success on the test partition is therefore a single-cycle event that depends on the seed and on the composition of the training set, and it is not evidence that the oscillatory archetype generalises. The augmentation schemes examined here did not establish robust F06 recognition across independent units; they do not rule out benefits from other augmentation procedures. Without F06 augmentation, neither tuned model recognises the single F06 cycle of the test partition (it is predicted as F02 and F03, respectively), whereas with augmentation the confusion between F03 and F06 persists (Section 5.9); the first observation rests on one reused cycle. The auxiliary checks (S1.13) show that separability from F03 is archetype-specific and that holding SN 045 out also removes most N02 cycles. These findings

support collecting more independent units and archetypes, while Section 5.11 quantifies the calibration coverage required for a guarantee on critical misses.

S1.26. Excluded Earlier Analyses

This history documents the scope of the exclusions stated in Section 4.12.

Analyses obtained with an earlier configuration of the pipeline, whose code and feature definitions were not retained, are not reported: a 113-descriptor feature set (of which F115 is a reconstruction, not a copy) with the corresponding classifier-family comparison, hyperparameter searches for Random Forest and XGBoost, a six-class composite stratification, and impurity and permutation importances on the F03/F06 subset.

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